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(54) **ARTIFICIAL INTELLIGENCE TRAINING USING ACCESIBILITY DATA**

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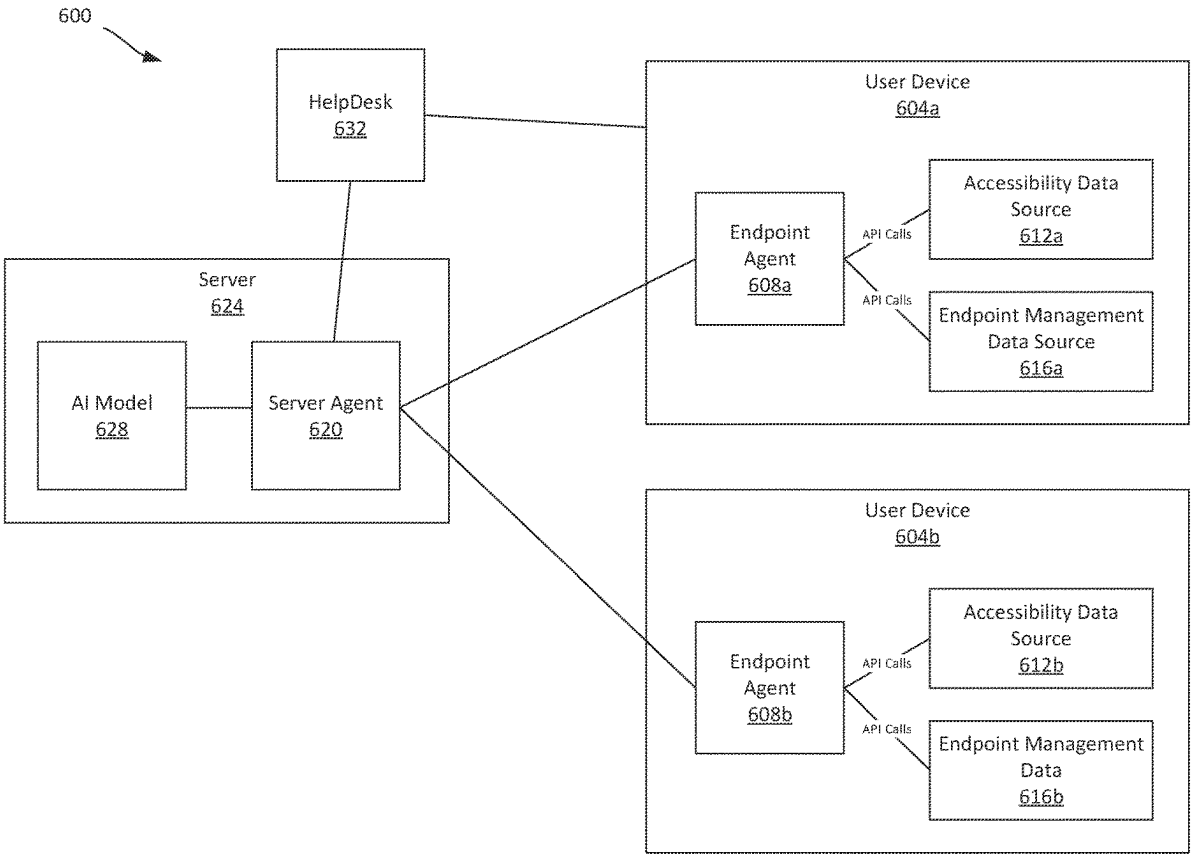
(57) **ABSTRACT**

Examples provide an electronic device including at least one electronic processor configured to request first accessibility data associated with a first user interface ("UI") displayed by a first device and including at least one of (a) information identifying one or more UI elements in the first UI or (b) information identifying one or more UI events in the first UI; train, based on at least the first accessibility data and an error state associated with the first device, an artificial intelligence ("AI") model; request second accessibility data associated with a second UI displayed by a second device and including (a) information identifying one or more UI elements in the second UI or (b) information identifying one or more UI events in the second UI; and detect, based on at least the second accessibility data and the AI model, the error state on the second device.



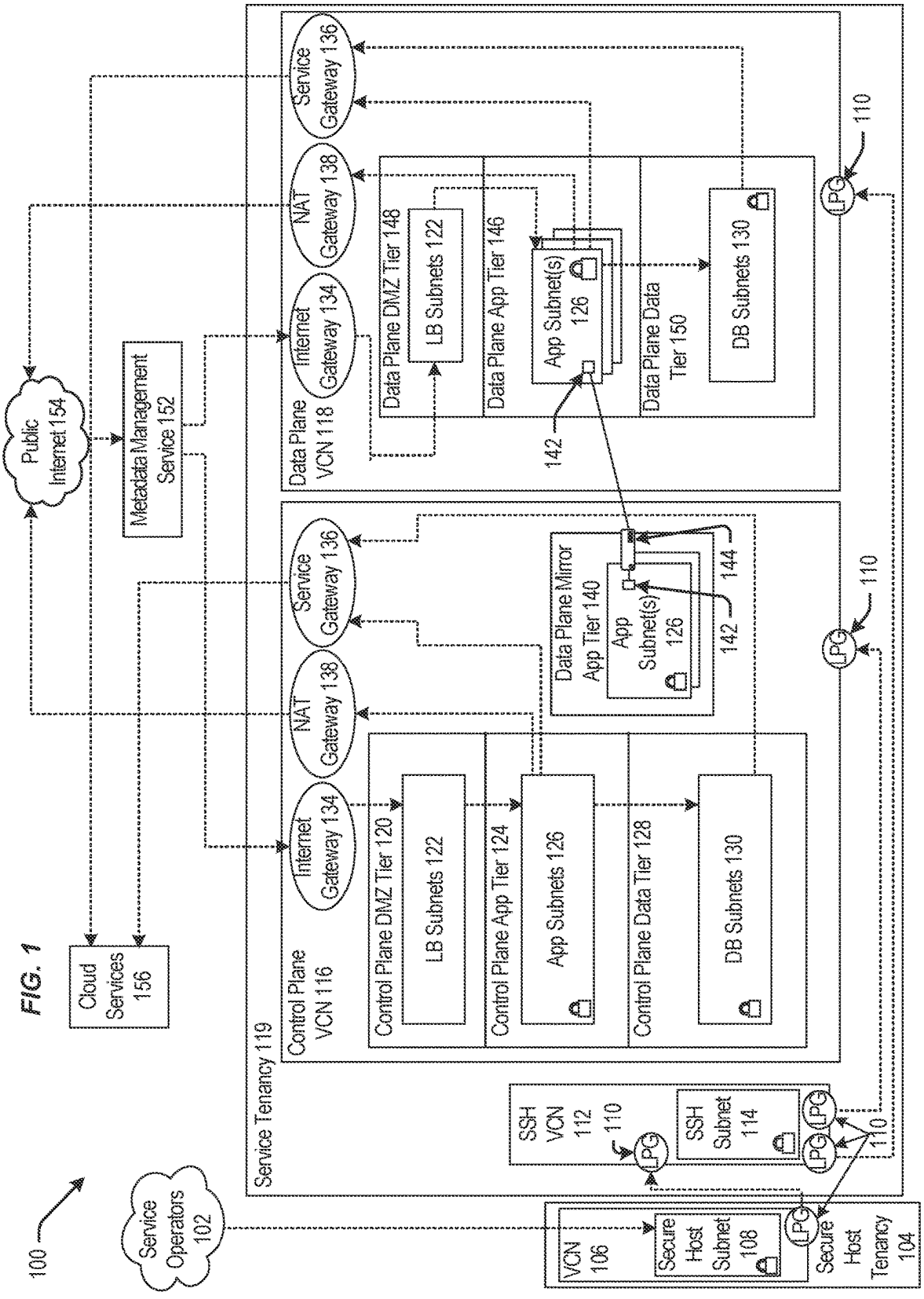
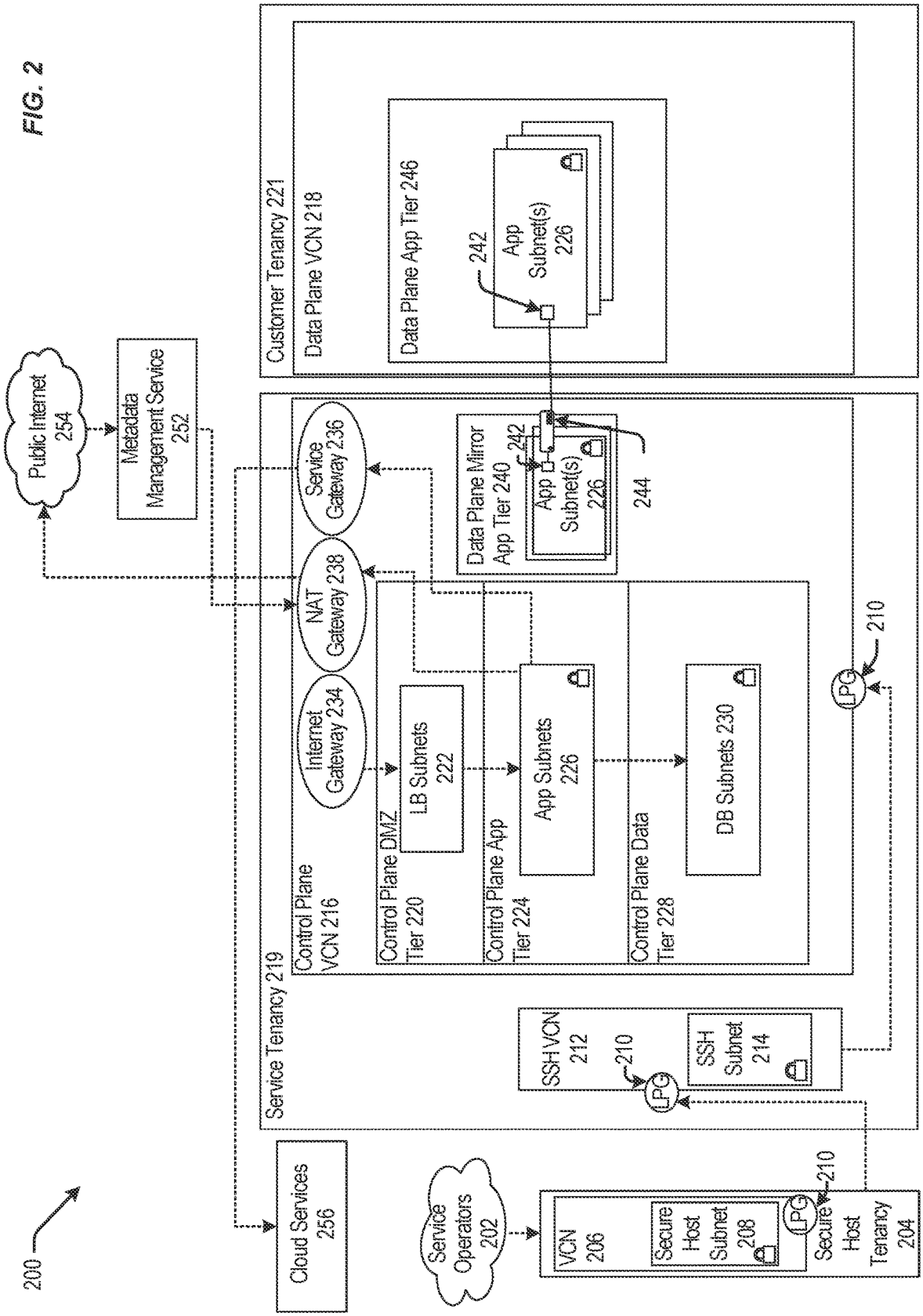

100
Service Operators 102
Secure Host Tenancy 104
VCN 106
Secure Host Subnet 108
SSH VCN 112
SSH Subnet 114
Service Tenancy 119
Control Plane VCN 116
Control Plane DMZ Tier 120
LB Subnets 122
Control Plane App Tier 124
App Subnets 126
Control Plane Data Tier 128
DB Subnets 130
Internet Gateway 134
NAT Gateway 138
Service Gateway 136
Data Plane Mirror App Tier 140
App Subnet(s) 126
142
144
Data Plane VCN 118
Data Plane DMZ Tier 148
LB Subnets 122
Data Plane App Tier 146
App Subnet(s) 126
142
Data Plane Data Tier 150
DB Subnets 130
Internet Gateway 134
NAT Gateway 138
Service Gateway 136
LPG
110
Public Internet 154
Metadata Management Service 152
Cloud Services 156

FIG. 1

FIG. 2


200
Public Internet 254
Metadata Management Service 252
Customer Tenancy 221
Data Plane VCN 218
Data Plane App Tier 246
242
App Subnet(s) 226
Service Tenancy 219
Control Plane VCN 216
Service Gateway 236
NAT Gateway 238
Internet Gateway 234
Data Plane Mirror App Tier 240
242
App Subnet(s) 226
244
Control Plane DMZ Tier 220
LB Subnets 222
Control Plane App Tier 224
App Subnets 226
Control Plane Data Tier 228
DB Subnets 230
210
LPG
SSH VCN 212
210
LPG
SSH Subnet 214
Cloud Services 256
Service Operators 202
VCN 206
Secure Host Subnet 208
LPG
210
Secure Host Tenancy 204

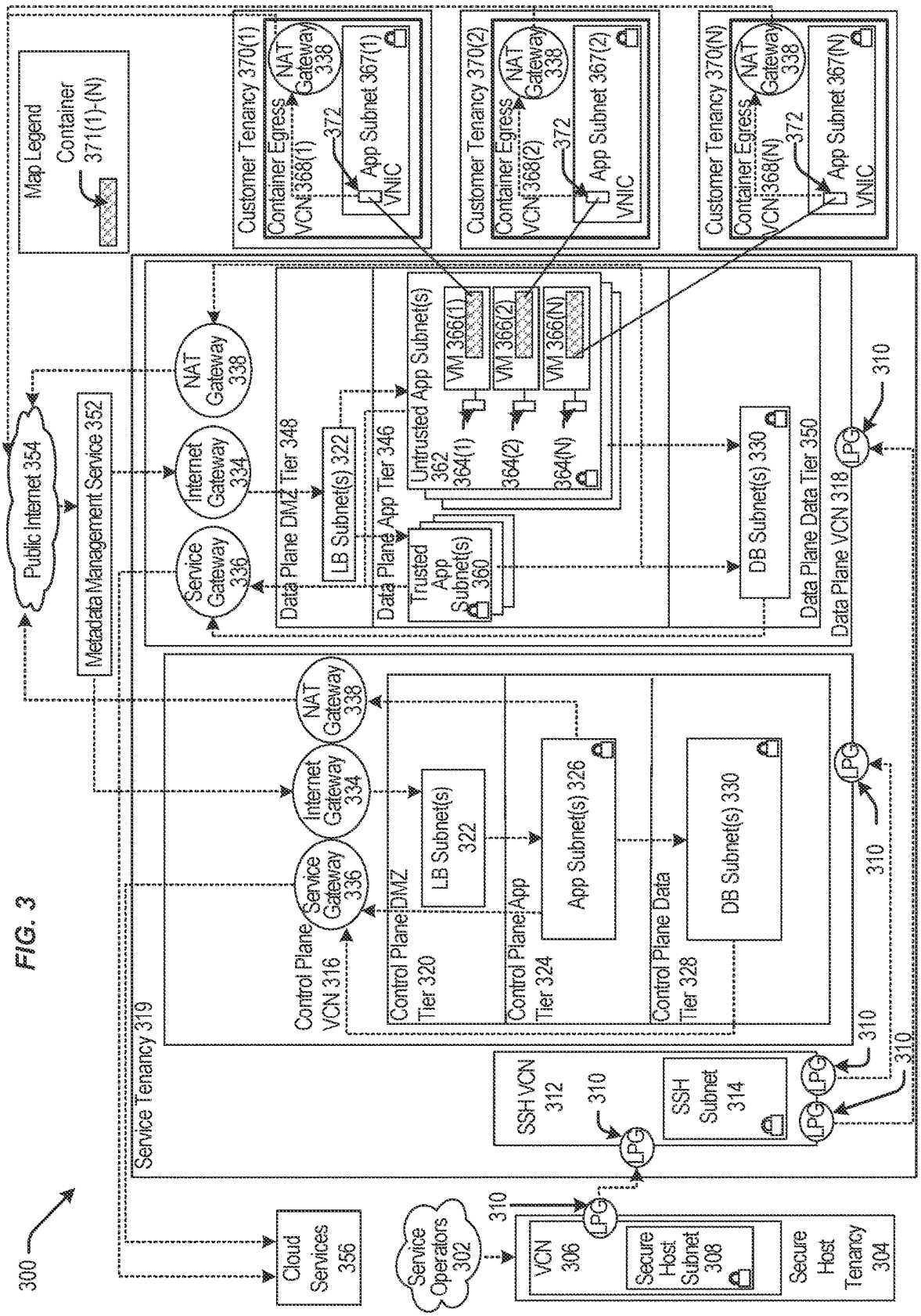

300
Map Legend
Container
371(1)-(N)
Public Internet 354
Metadata Management Service 352
Cloud Services 356
Service Operators 302
310
VCN 306
Secure Host Subnet 308
Secure Host Tenancy 304
SSH VCN 312
SSH Subnet 314
Service Tenancy 319
Control Plane VCN 316
Control Plane DMZ Tier 320
LB Subnet(s) 322
Control Plane App Tier 324
App Subnet(s) 326
Control Plane Data Tier 328
DB Subnet(s) 330
Service Gateway 336
Internet Gateway 334
NAT Gateway 338
Data Plane VCN 318
Data Plane DMZ Tier 348
Data Plane App Tier 346
Data Plane Data Tier 350
Untrusted App Subnet(s) 362
Trusted App Subnet(s) 360
VM 366(1)
VM 366(2)
VM 366(N)
364(1)
364(2)
364(N)
Customer Tenancy 370(1)
Container Egress VCN 368(1)
372
App Subnet 367(1)
VNIC
Customer Tenancy 370(2)
Container Egress VCN 368(2)
App Subnet 367(2)
Customer Tenancy 370(N)
Container Egress VCN 368(N)
App Subnet 367(N)
LPG

FIG. 3

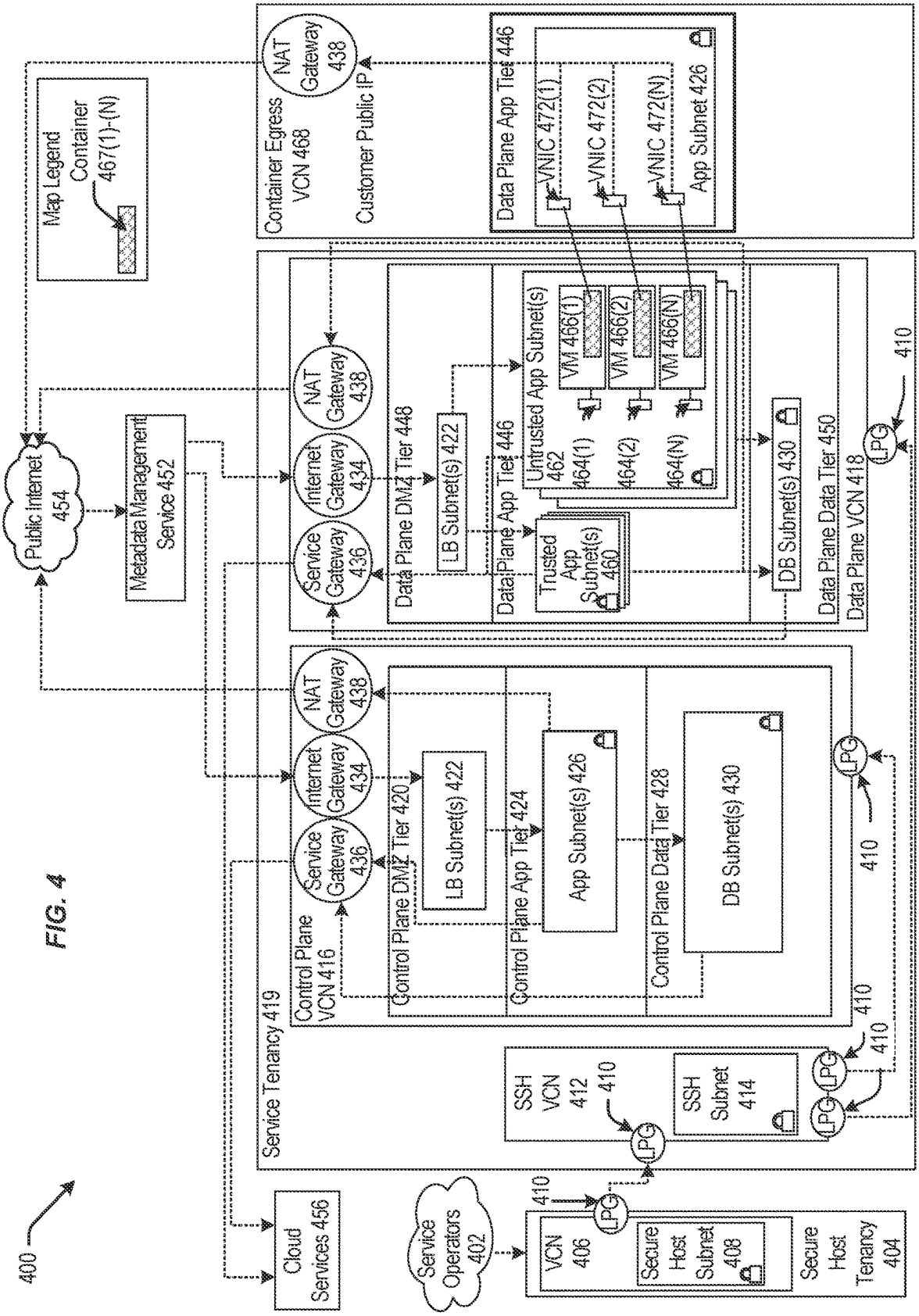

400
Map Legend
Container
467(1)-(N)
Container Egress VCN 468
NAT Gateway 438
Customer Public IP
Data Plane App Tier 446
VNIC 472(1)
VNIC 472(2)
VNIC 472(N)
App Subnet 426
Public Internet 454
Metadata Management Service 452
Service Gateway 436
Internet Gateway 434
NAT Gateway 438
Data Plane DMZ Tier 448
LB Subnet(s) 422
Data Plane App Tier 446
Untrusted App Subnet(s) 462
VM 466(1)
VM 466(2)
VM 466(N)
464(1)
464(2)
464(N)
Trusted App Subnet(s) 460
DB Subnet(s) 430
Data Plane Data Tier 450
Data Plane VCN 418
410
Service Tenancy 419
Control Plane VCN 416
Control Plane DMZ Tier 420
LB Subnet(s) 422
Control Plane App Tier 424
App Subnet(s) 426
Control Plane Data Tier 428
DB Subnet(s) 430
SSH VCN 412
SSH Subnet 414
LPG
Cloud Services 456
Service Operators 402
VCN 406
Secure Host Subnet 408
Secure Host Tenancy 404

FIG. 4

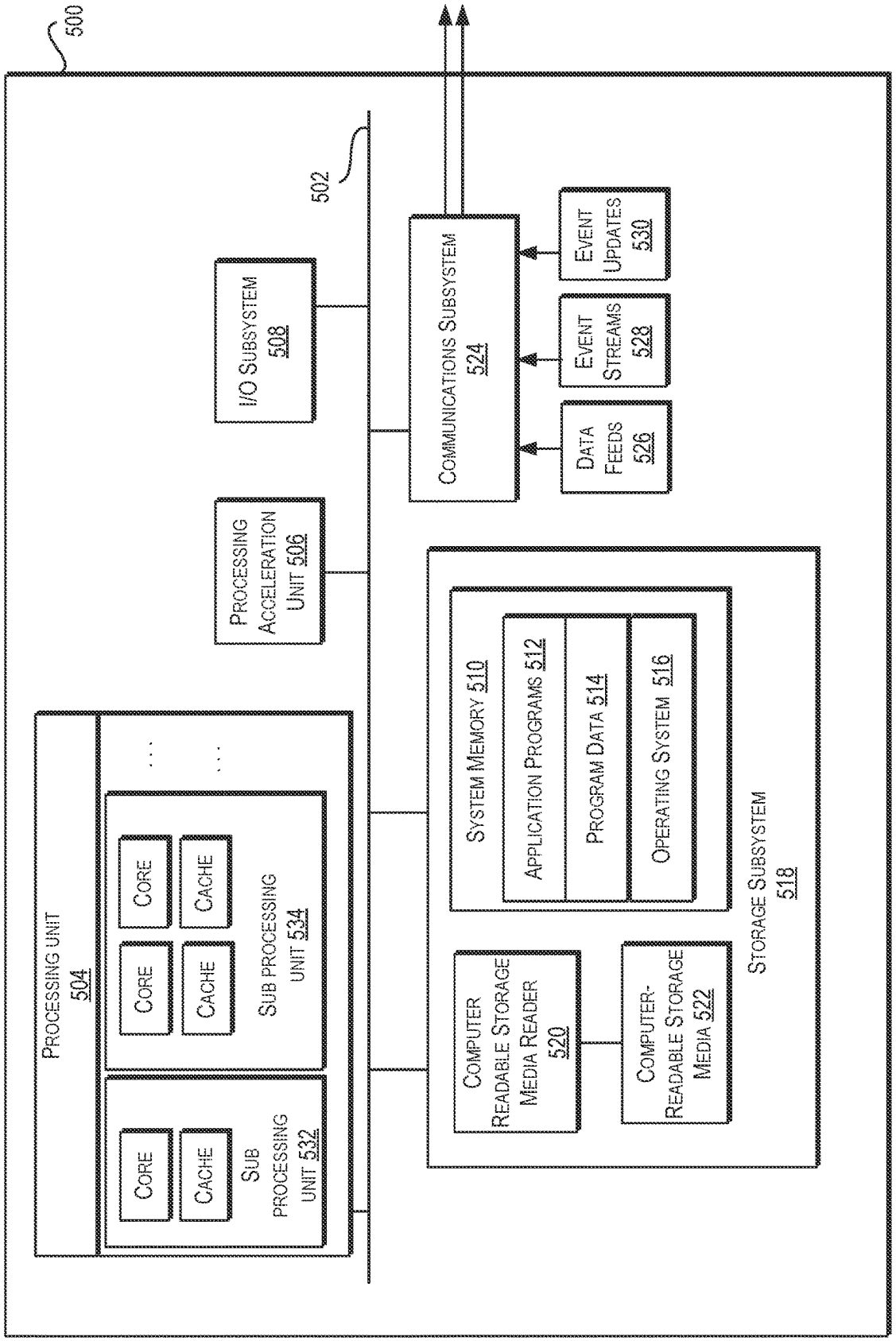

500
502
PROCESSING UNIT 504
CORE
CACHE
SUB PROCESSING UNIT 532
CORE
CACHE
SUB PROCESSING UNIT 534
. . .
. . .
PROCESSING ACCELERATION UNIT 506
I/O SUBSYSTEM 508
COMMUNICATIONS SUBSYSTEM 524
DATA FEEDS 526
EVENT STREAMS 528
EVENT UPDATES 530
STORAGE SUBSYSTEM 518
SYSTEM MEMORY 510
APPLICATION PROGRAMS 512
PROGRAM DATA 514
OPERATING SYSTEM 516
COMPUTER READABLE STORAGE MEDIA READER 520
COMPUTER-READABLE STORAGE MEDIA 522

FIG. 5

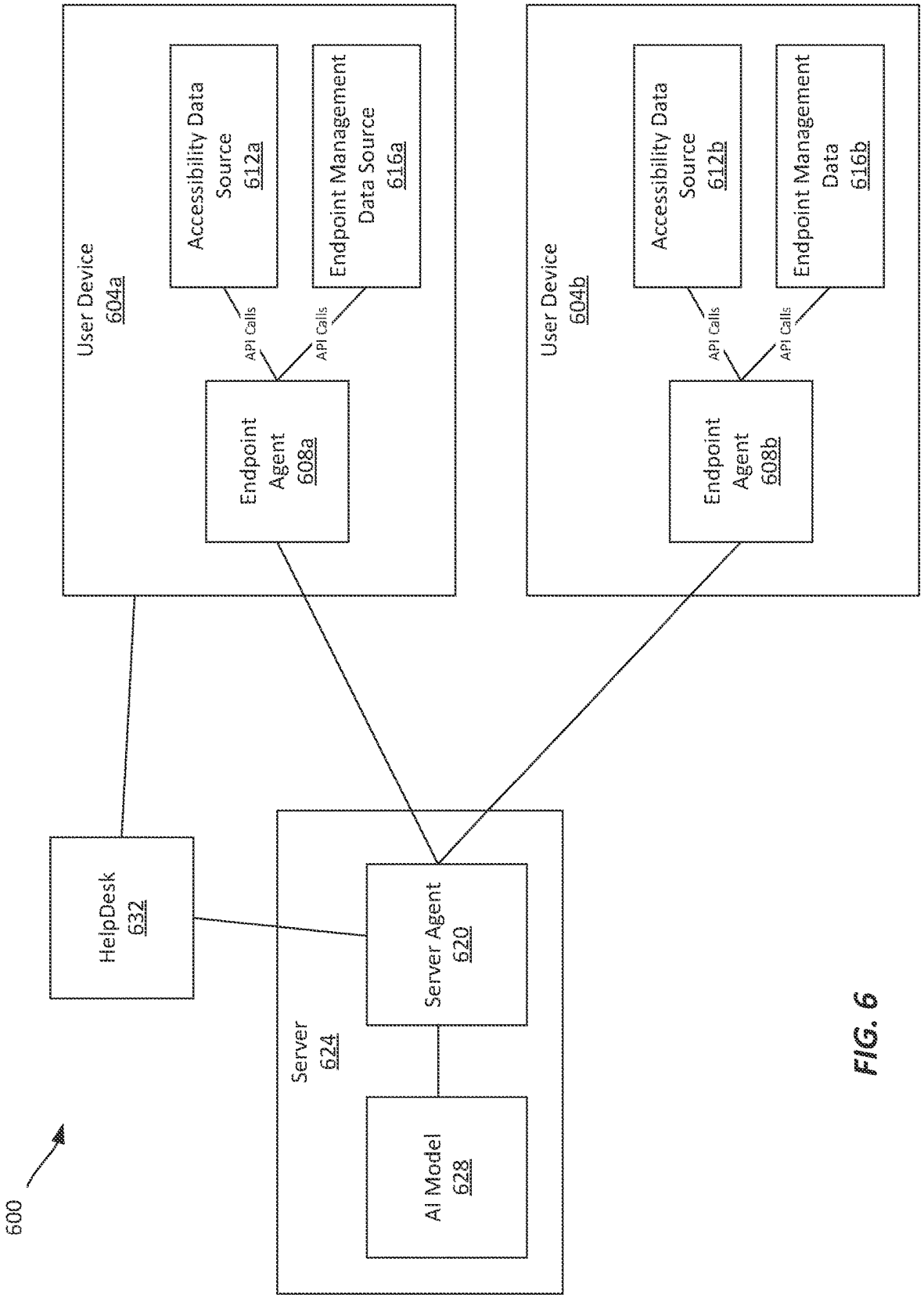

600
User Device
604a
Accessibility Data Source
612a
Endpoint Management Data Source
616a
API Calls
API Calls
Endpoint Agent
608a
User Device
604b
Accessibility Data Source
612b
Endpoint Management Data
616b
API Calls
API Calls
Endpoint Agent
608b
HelpDesk
632
Server
624
Server Agent
620
AI Model
628

FIG. 6

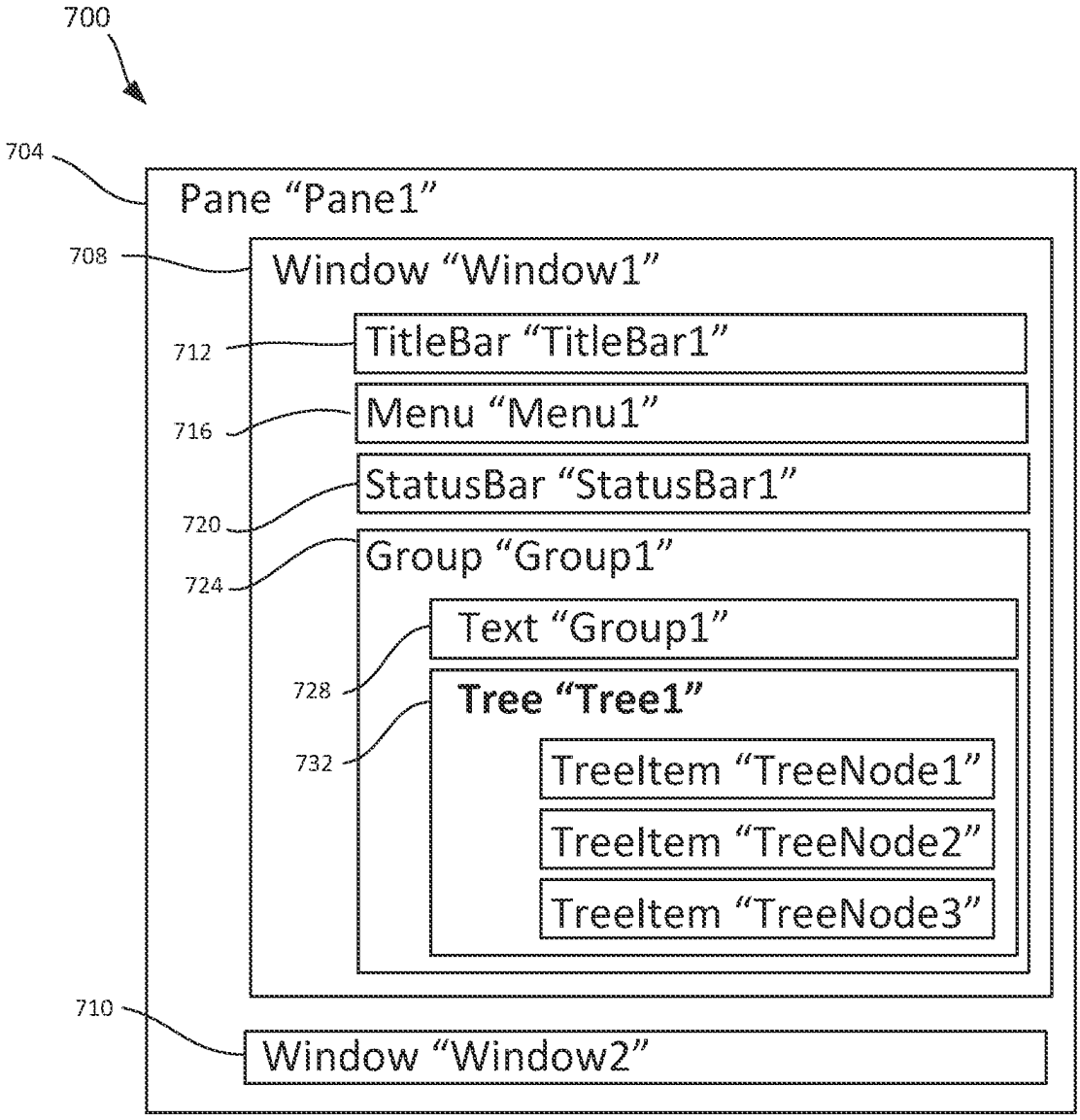

700
704
Pane "Pane1"
708
Window "Window1"
712
TitleBar "TitleBar1"
716
Menu "Menu1"
StatusBar "StatusBar1"
720
Group "Group1"
724
Text "Group1"
728
Tree "Tree1"
732
TreeItem "TreeNode1"
TreeItem "TreeNode2"
TreeItem "TreeNode3"
710
Window "Window2"

FIG. 7A

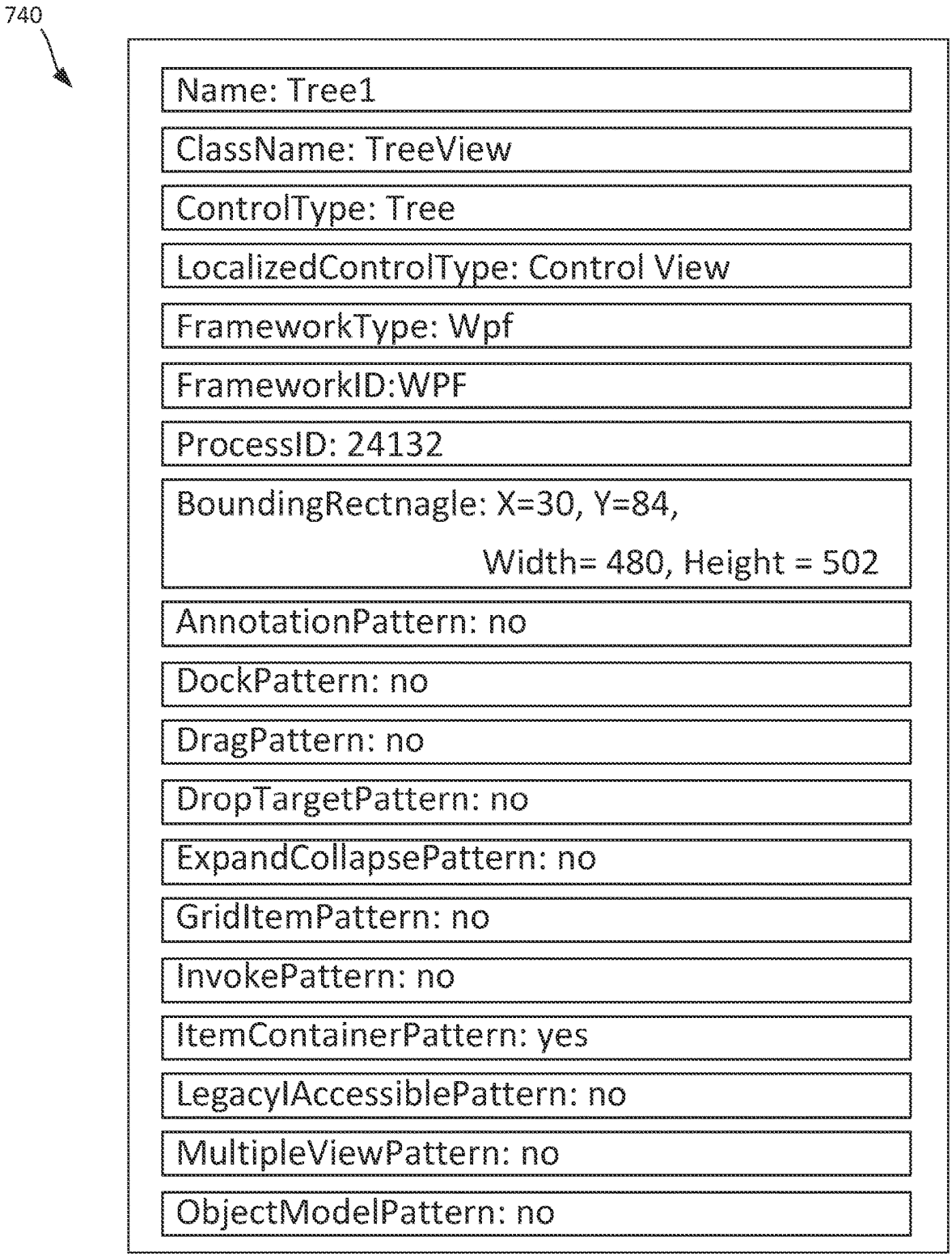

740
Name: Tree1
ClassName: TreeView
ControlType: Tree
LocalizedControlType: Control View
FrameworkType: Wpf
FrameworkID:WPF
ProcessID: 24132
BoundingRectnagle: X=30, Y=84,
Width= 480, Height = 502
AnnotationPattern: no
DockPattern: no
DragPattern: no
DropTargetPattern: no
ExpandCollapsePattern: no
GridItemPattern: no
InvokePattern: no
ItemContainerPattern: yes
LegacyIAccessiblePattern: no
MultipleViewPattern: no
ObjectModelPattern: no

FIG. 7B

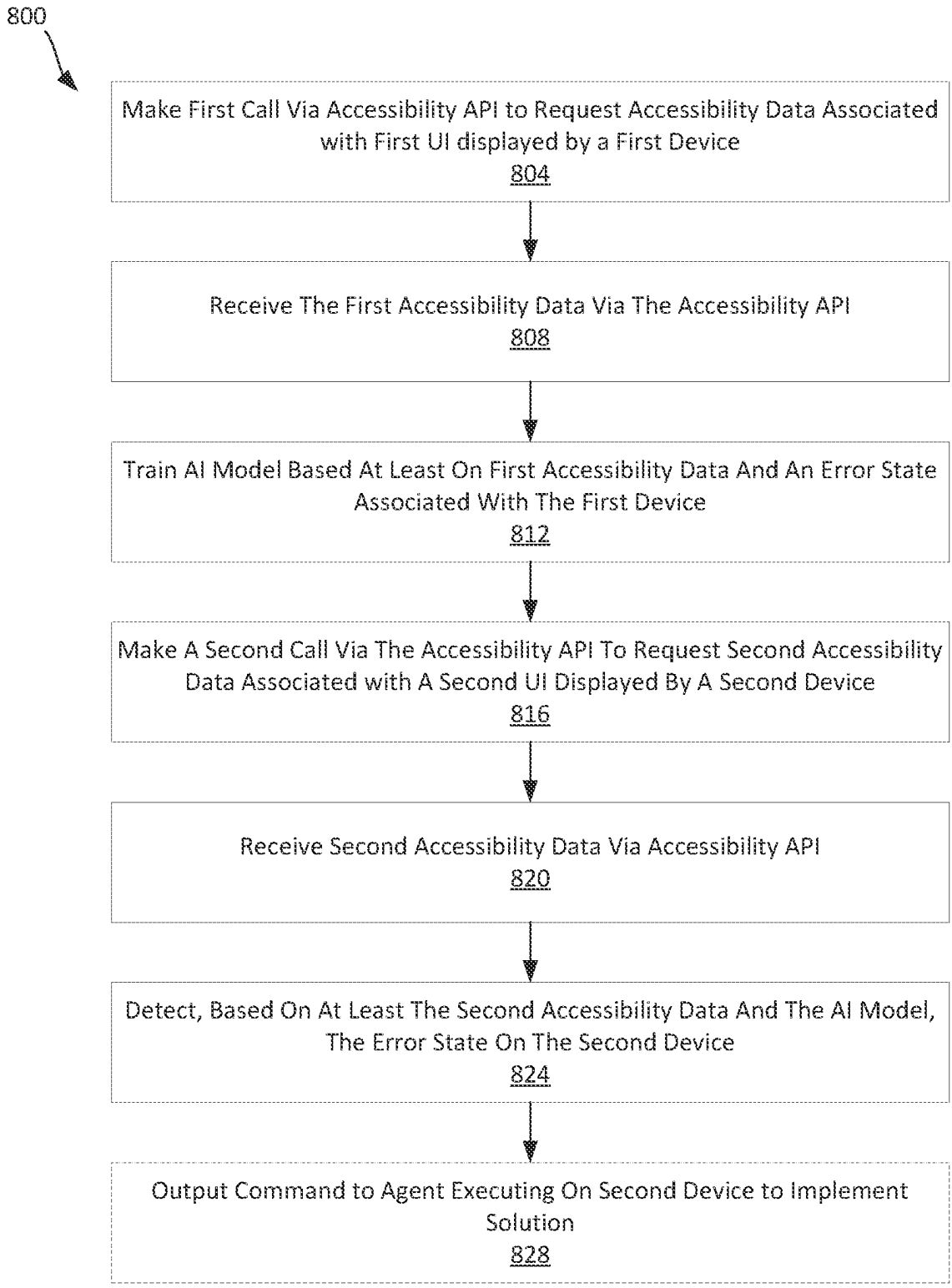

800
Make First Call Via Accessibility API to Request Accessibility Data Associated with First UI displayed by a First Device
804
Receive The First Accessibility Data Via The Accessibility API
808
Train AI Model Based At Least On First Accessibility Data And An Error State Associated With The First Device
812
Make A Second Call Via The Accessibility API To Request Second Accessibility Data Associated with A Second UI Displayed By A Second Device
816
Receive Second Accessibility Data Via Accessibility API
820
Detect, Based On At Least The Second Accessibility Data And The AI Model, The Error State On The Second Device
824
Output Command to Agent Executing On Second Device to Implement Solution
828

FIG. 8

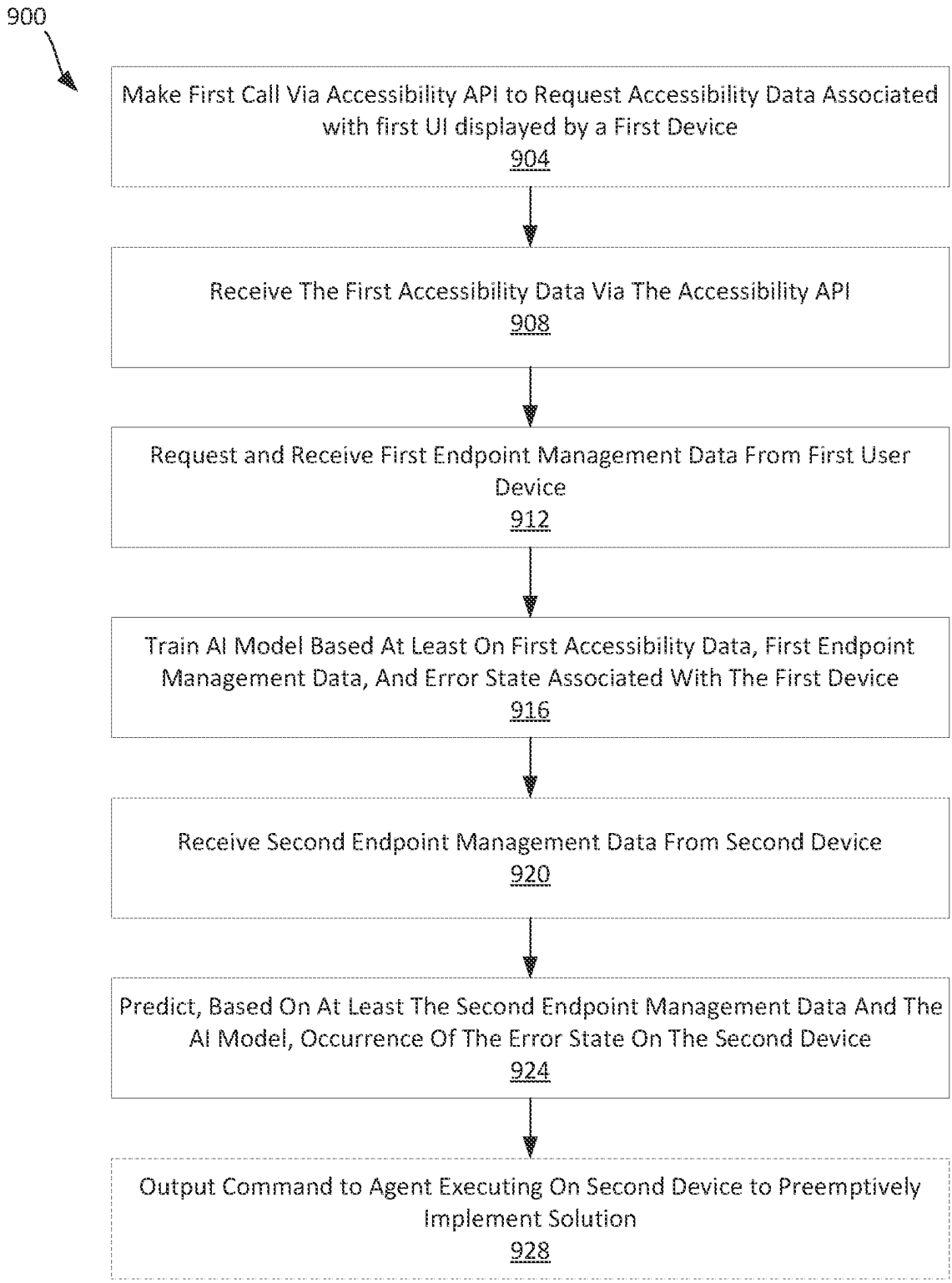

900
Make First Call Via Accessibility API to Request Accessibility Data Associated with first UI displayed by a First Device
904
Receive The First Accessibility Data Via The Accessibility API
908
Request and Receive First Endpoint Management Data From First User Device
912
Train AI Model Based At Least On First Accessibility Data, First Endpoint Management Data, And Error State Associated With The First Device
916
Receive Second Endpoint Management Data From Second Device
920
Predict, Based On At Least The Second Endpoint Management Data And The AI Model, Occurrence Of The Error State On The Second Device
924
Output Command to Agent Executing On Second Device to Preemptively Implement Solution
928

FIG. 9

ARTIFICIAL INTELLIGENCE TRAINING USING ACCESIBILITY DATA

FIELD

[0001] Embodiments described herein generally relate to training artificial intelligence models.

SUMMARY

[0002] Conventional helpdesk bots integrated with a helpdesk service used for troubleshooting errors in software applications are trained only on text-based diagnostics provided directly by an end-user, and, as a result, are often ineffective at identifying and solving issues that the end-user experiences. For example, an end-user experiencing an error state on an end-user device may not know what information to provide a helpdesk bot, and the helpdesk bot may lack access to operational data associated with the endpoint device. These conventional helpdesk bots are therefore only marginally improved over coded troubleshooting trees. As a result, users in an organization often rely on a manual helpdesk process to troubleshoot errors, which may lack the resources to efficiently troubleshoot errors. Additionally, such helpdesks are typically only used to troubleshoot and solve errors after errors have occurred, rather than prevent the future occurrence of such errors.

[0003] Thus, there is a need for improved troubleshooting of endpoint devices using an artificial intelligence ("AI") model trained on near real-time data from, for example, an accessibility interface in conjunction with telemetry, logs, and state information provided by an endpoint agent to learn, troubleshoot, resolve, and validate resolutions to endpoint device error states. This solution allows the AI model to perform new approaches to troubleshooting, modify solutions, and implement solutions to multiple devices within an organization—even before all such devices experience an error.

[0004] For example, one example provides an electronic device including at least one electronic processor configured to: make a first call via an accessibility application programming interface ("API") to request first accessibility data associated with a first user interface ("UI") displayed by a first device; receive the first accessibility data via the accessibility API, the first accessibility data comprising at least one of (a) information identifying one or more UI elements in the first UI or (b) information identifying one or more UI events in the first UI; train, based on at least the first accessibility data and an error state associated with the first device, an artificial intelligence ("AI") model; make a second call via the accessibility API to request second accessibility data associated with a second UI displayed by a second device; receive the second accessibility data via the accessibility API, the second accessibility data comprising (a) information identifying one or more UI elements in the second UI or (b) information identifying one or more UI events in the second UI; and detect, based on at least the second accessibility data and the AI model, the error state on the second device.

[0005] In some aspects, the error state of the first device is reported through a helpdesk service.

[0006] In some aspects, the at least one electronic processor is further configured to determine the error state of the first device based on at least first endpoint management data, the first endpoint management data including state information of the first device.

[0007] In some aspects, the AI model is further trained based on at least a solution to the error state, and wherein the at least one electronic processor is further configured to: output, based on at least the AI model, a command to an agent executing on the second device to implement the solution on the second device.

[0008] In some aspects, the solution includes at least one selected from a group consisting of restarting at least one application executing on the second device, modifying an application configuration of at least one application of the second device, and modifying a network configuration of the second device.

[0009] In some aspects, the AI model is further trained based on first endpoint management data and a solution to the error state, the first endpoint management data including state information of the first device, wherein the at least one electronic processor is further configured to: predict, based on at least the AI model and second endpoint management data including state information of a third device, an occurrence of the error state on the third device; and output, based on at least the AI model, a command to an agent executing on the third device to preemptively implement the solution on the third device.

[0010] In some aspects, the at least one electronic processor is further configured to: receive third endpoint management data after outputting the command, the third endpoint management device including state information of the third device; and train the AI model based on at least the third endpoint management data.

[0011] In some aspects, the information identifying the one or more UI elements in the first UI includes at least one selected from the group consisting of: (a) a respective element type of the one or more UI elements in the first UI, (b) a respective identifier of the one or more UI elements in the first UI, or (c) a respective state or condition of the one or more UI elements in the first UI.

[0012] In some aspects, the information identifying the one or more UI elements in the first UI is organized as a hierarchical tree.

[0013] In some aspects, the one or more UI elements in the first UI comprises a first UI element that includes a second UI element; the hierarchical tree comprises a first hierarchical level that is above a second hierarchical level; the first hierarchical level includes a first node representing the first UI element; and the second hierarchical level includes a second node representing the second UI element.

[0014] In some aspects, the information identifying the one or more UI events in the first UI includes one or more notifications of changes in states or conditions of the UI elements in the first UI.

[0015] In some aspects, the first call made via the accessibility API is made to a platform rendering the first UI.

[0016] In some aspects, the platform includes one of an operating system executing on the first device or a browser application executing on the first device.

[0017] In some aspects, the first accessibility data is generated based on metadata associated with the one or more UI elements in the first UI, the metadata being exposed via the accessibility API.

[0018] In some aspects, the metadata is specified in one of (a) application code of application for which the first UI is

being rendered or (b) content code of web content for which the first UI is being rendered.

[0019] In some aspects, the metadata specifies hierarchical relationships between the one or more UI elements in the first UI.

[0020] In some aspects, the first accessibility data is generated by a platform executing on the first device during rendering of the first UI.

[0021] Another example provides a method for training an artificial intelligence ("AI") model for error troubleshooting. The method includes making a first call via an accessibility application programming interface ("API") to request first accessibility data associated with a first user interface ("UI") displayed by a first device; receiving the first accessibility data via the accessibility API, the first accessibility data comprising at least one of (a) information identifying one or more UI elements in the first UI or (b) information identifying one or more UI events in the first UI; training, based on at least the first accessibility data and an error state associated with the first device, the AI model; making a second call via the accessibility API to request second accessibility data associated with a second UI displayed by a second device; receiving the second accessibility data via the accessibility API, the second accessibility data comprising (a) information identifying one or more UI elements in the second UI or (b) information identifying one or more UI events in the second UI; and detecting, based on at least the second accessibility data and the AI model, the error state on the second device.

[0022] In some aspects, the method further includes training the AI model based on at least a solution to the error state; and outputting, based on at least the AI model, a command to an agent executing on the second device to implement the solution on the second device.

[0023] Another example provides an electronic device including at least one electronic processor configured to: make a first call via an accessibility application programming interface ("API") to request first accessibility data associated with a first user interface ("UI") displayed by a first device; receive the first accessibility data via the accessibility API, the first accessibility data comprising at least one of (a) information identifying one or more UI elements in the first UI or (b) information identifying one or more UI events in the first UI; train an artificial intelligence ("AI") model based on at least the first accessibility data, an error state associated with the first device, and endpoint management data associated with the first device, the endpoint management data including state information of the first device; and predict, based on at least the AI model and second endpoint management data including state information of a second device, an occurrence of the error state on the second device.

[0024] Other aspects will become apparent by consideration of the detailed description and accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] FIG. **1** is a block diagram illustrating one pattern for implementing a cloud infrastructure as a service system, according to at least one embodiment.

[0026] FIG. **2** is a block diagram illustrating another pattern for implementing a cloud infrastructure as a service system, according to at least one embodiment.

[0027] FIG. **3** is a block diagram illustrating another pattern for implementing a cloud infrastructure as a service system, according to at least one embodiment.

[0028] FIG. **4** is a block diagram illustrating another pattern for implementing a cloud infrastructure as a service system, according to at least one embodiment.

[0029] FIG. **5** is a block diagram illustrating an example computer system, according to at least one embodiment.

[0030] FIG. **6** is a block diagram illustrating a system architecture for troubleshooting errors, according to at least one embodiment.

[0031] FIG. **7**A illustrates an example accessibility data tree, according to at least one embodiment.

[0032] FIG. **7**B illustrates an example details portion of a selected UI element of the accessibility data tree of FIG. **7**A, according to at least one embodiment.

[0033] FIG. **8** is a flowchart illustrating an example method for training an artificial intelligence model to detect error states, according to at least one embodiment.

[0034] FIG. **9** is a flowchart illustrating an example method for training an artificial intelligence model to predict error states, according to at least one embodiment.

DETAILED DESCRIPTION

[0035] In the following description, various embodiments will be described. For purposes of explanation, specific configurations and details are set forth to provide a thorough understanding of the embodiments. However, it will also be apparent to one skilled in the art that the embodiments may be practiced without the specific details. Furthermore, well-known features may be omitted or simplified in order not to obscure the embodiment being described.

Cloud-Based Computing Platforms

[0036] Embodiments described herein may performed, wholly or partly, within a cloud-based computing platform. Cloud-based computing platforms provide scalable and flexible computing resources for users. Infrastructure as a service (IaaS) is one particular type of cloud computing. IaaS can be configured to provide virtualized computing resources over a public network (e.g., the Internet). In an IaaS model, a cloud computing provider can host the infrastructure components (e.g., servers, storage devices, network nodes (e.g., hardware), deployment software, platform virtualization (e.g., a hypervisor layer), or the like). In some cases, an IaaS provider may also supply a variety of services to accompany those infrastructure components (example services include billing software, monitoring software, logging software, load balancing software, clustering software, etc.). Thus, as these services may be policy-driven, IaaS users may be able to implement policies to drive load balancing to maintain application availability and performance.

[0037] In some instances, IaaS customers may access resources and services through a wide area network (WAN), such as the Internet, and can use the cloud provider's services to install the remaining elements of an application stack. For example, the user can log in to the IaaS platform to create virtual machines (VMs), install operating systems (OSs) on each VM, deploy middleware such as databases, create storage buckets for workloads and backups, and even install enterprise software into that VM. Customers can then use the provider's services to perform various functions,

including balancing network traffic, troubleshooting application issues, monitoring performance, managing disaster recovery, etc.

[0038] In most cases, a cloud computing model will require the participation of a cloud provider. The cloud provider may, but need not be, a third-party service that specializes in providing (e.g., offering, renting, selling) IaaS. An entity might also opt to deploy a private cloud, becoming its own provider of infrastructure services.

[0039] In some examples, IaaS deployment is the process of putting a new application, or a new version of an application, onto a prepared application server or the like. It may also include the process of preparing the server (e.g., installing libraries, daemons, etc.). This is often managed by the cloud provider, below the hypervisor layer (e.g., the servers, storage, network hardware, and virtualization). Thus, the customer may be responsible for handling (OS), middleware, and/or application deployment (e.g., on self-service virtual machines (e.g., that can be spun up on demand) or the like.

[0040] In some examples, IaaS provisioning may refer to acquiring computers or virtual hosts for use, and even installing needed libraries or services on them. In most cases, deployment does not include provisioning, and the provisioning may need to be performed first.

[0041] In some cases, there are two different challenges for IaaS provisioning. First, there is the initial challenge of provisioning the initial set of infrastructure before anything is running. Second, there is the challenge of evolving the existing infrastructure (e.g., adding new services, changing services, removing services, etc.) once everything has been provisioned. In some cases, these two challenges may be addressed by enabling the configuration of the infrastructure to be defined declaratively. In other words, the infrastructure (e.g., what components are needed and how they interact) can be defined by one or more configuration files. Thus, the overall topology of the infrastructure (e.g., what resources depend on which, and how they each work together) can be described declaratively. In some instances, once the topology is defined, a workflow can be generated that creates and/or manages the different components described in the configuration files.

[0042] In some examples, an infrastructure may have many interconnected elements. For example, there may be one or more virtual private clouds (VPCs) (e.g., a potentially on-demand pool of configurable and/or shared computing resources), also known as a core network. In some examples, there may also be one or more inbound/outbound traffic group rules provisioned to define how the inbound and/or outbound traffic of the network will be set up and one or more virtual machines (VMs). Other infrastructure elements may also be provisioned, such as a load balancer, a database, or the like. As more and more infrastructure elements are desired and/or added, the infrastructure may incrementally evolve.

[0043] In some instances, continuous deployment techniques may be employed to enable deployment of infrastructure code across various virtual computing environments. Additionally, the described techniques can enable infrastructure management within these environments. In some examples, service teams can write code that is desired to be deployed to one or more, but often many, different production environments (e.g., across various different geographic locations, sometimes spanning the entire world). However, in some examples, the infrastructure on which the code will be deployed must first be set up. In some instances, the provisioning can be done manually, a provisioning tool may be utilized to provision the resources, and/or deployment tools may be utilized to deploy the code once the infrastructure is provisioned.

[0044] FIG. **1** is a block diagram **100** illustrating an example pattern of an IaaS architecture, according to at least one embodiment. Service operators **102** can be communicatively coupled to a secure host tenancy **104** that can include a virtual cloud network (VCN) **106** and a secure host subnet **108**. In some examples, the service operators **102** may use one or more client computing devices, which may be portable handheld devices (e.g., an iPhone®, cellular telephone, an iPad®, computing tablet, a personal digital assistant (PDA)) or wearable devices (e.g., a Google Glass® head mounted display), running software and/or a variety of mobile operating systems such as iOS, Windows Phone, Android, BlackBerry 8, Palm OS, and the like, and being Internet, e-mail, short message service (SMS), Blackberry®, or other communication protocol enabled. Alternatively, the client computing devices can be general purpose personal computers including, by way of example, personal computers and/or laptop computers running various versions of Microsoft Windows®, Apple Macintosh®, and/or Linux operating systems. The client computing devices can be workstation computers running any of a variety of commercially-available UNIX® or UNIX-like operating systems, including without limitation the variety of GNU/Linux operating systems, such as for example, Google Chrome OS. Alternatively, or in addition, client computing devices may be any other electronic device, such as a thin-client computer, an Internet-enabled gaming system (e.g., a Microsoft Xbox gaming console with or without a Kinect® gesture input device), and/or a personal messaging device, capable of communicating over a network that can access the VCN **106** and/or the Internet.

[0045] The VCN **106** can include a local peering gateway (LPG) **110** that can be communicatively coupled to a secure shell (SSH) VCN **112** via an LPG **110** contained in the SSH VCN **112**. The SSH VCN **112** can include an SSH subnet **114**, and the SSH VCN **112** can be communicatively coupled to a control plane VCN **116** via the LPG **110** contained in the control plane VCN **116**. Also, the SSH VCN **112** can be communicatively coupled to a data plane VCN **118** via an LPG **110**. The control plane VCN **116** and the data plane VCN **118** can be contained in a service tenancy **119** that can be owned and/or operated by the IaaS provider.

[0046] The control plane VCN **116** can include a control plane demilitarized zone (DMZ) tier **120** that acts as a perimeter network (e.g., portions of a corporate network between the corporate intranet and external networks). The DMZ-based servers may have restricted responsibilities and help keep breaches contained. Additionally, the DMZ tier **120** can include one or more load balancer (LB) subnet(s) **122**, a control plane app tier **124** that can include app subnet(s) **126**, a control plane data tier **128** that can include database (DB) subnet(s) **130** (e.g., frontend DB subnet(s) and/or backend DB subnet(s)). The LB subnet(s) **122** contained in the control plane DMZ tier **120** can be communicatively coupled to the app subnet(s) **126** contained in the control plane app tier **124** and an Internet gateway **134** that can be contained in the control plane VCN **116**, and the app subnet(s) **126** can be communicatively coupled to the DB

subnet(s) **130** contained in the control plane data tier **128** and a service gateway **136** and a network address translation (NAT) gateway **138**. The control plane VCN **116** can include the service gateway **136** and the NAT gateway **138**.

[0047] The control plane VCN **116** can include a data plane mirror app tier **140** that can include app subnet(s) **126**. The app subnet(s) **126** contained in the data plane mirror app tier **140** can include a virtual network interface controller (VNIC) **142** that can execute a compute instance **144**. The compute instance **144** can communicatively couple the app subnet(s) **126** of the data plane mirror app tier **140** to app subnet(s) **126** that can be contained in a data plane app tier **146**.

[0048] The data plane VCN **118** can include the data plane app tier **146**, a data plane DMZ tier **148**, and a data plane data tier **150**. The data plane DMZ tier **148** can include LB subnet(s) **122** that can be communicatively coupled to the app subnet(s) **126** of the data plane app tier **146** and the Internet gateway **134** of the data plane VCN **118**. The app subnet(s) **126** can be communicatively coupled to the service gateway **136** of the data plane VCN **118** and the NAT gateway **138** of the data plane VCN **118**. The data plane data tier **150** can also include the DB subnet(s) **130** that can be communicatively coupled to the app subnet(s) **126** of the data plane app tier **146**.

[0049] The Internet gateway **134** of the control plane VCN **116** and of the data plane VCN **118** can be communicatively coupled to a metadata management service **152** that can be communicatively coupled to public Internet **154**. Public Internet **154** can be communicatively coupled to the NAT gateway **138** of the control plane VCN **116** and of the data plane VCN **118**. The service gateway **136** of the control plane VCN **116** and of the data plane VCN **118** can be communicatively coupled to cloud services **156**.

[0050] In some examples, the service gateway **136** of the control plane VCN **116** or of the data plane VCN **118** can make application programming interface (API) calls to cloud services **156** without going through public Internet **154**. The API calls to cloud services **156** from the service gateway **136** can be one-way: the service gateway **136** can make API calls to cloud services **156**, and cloud services **156** can send requested data to the service gateway **136**. But, cloud services **156** may not initiate API calls to the service gateway **136**.

[0051] In some examples, the secure host tenancy **104** can be directly connected to the service tenancy **119**, which may be otherwise isolated. The secure host subnet **108** can communicate with the SSH subnet **114** through an LPG **110** that may enable two-way communication over an otherwise isolated system. Connecting the secure host subnet **108** to the SSH subnet **114** may give the secure host subnet **108** access to other entities within the service tenancy **119**.

[0052] The control plane VCN **116** may allow users of the service tenancy **119** to set up or otherwise provision desired resources. Desired resources provisioned in the control plane VCN **116** may be deployed or otherwise used in the data plane VCN **118**. In some examples, the control plane VCN **116** can be isolated from the data plane VCN **118**, and the data plane mirror app tier **140** of the control plane VCN **116** can communicate with the data plane app tier **146** of the data plane VCN **118** via VNICs **142** that can be contained in the data plane mirror app tier **140** and the data plane app tier **146**.

[0053] In some examples, users of the system can make requests, for example create, read, update, or delete (CRUD) operations, through public Internet **154** that can communicate the requests to the metadata management service **152**. The metadata management service **152** can communicate the request to the control plane VCN **116** through the Internet gateway **134**. The request can be received by the LB subnet(s) **122** contained in the control plane DMZ tier **120**. The LB subnet(s) **122** may determine that the request is valid, and in response to this determination, the LB subnet(s) **122** can transmit the request to app subnet(s) **126** contained in the control plane app tier **124**. If the request is validated and requires a call to public Internet **154**, the call to public Internet **154** may be transmitted to the NAT gateway **138** that can make the call to public Internet **154**. Metadata that may be desired to be stored by the request can be stored in the DB subnet(s) **130**.

[0054] In some examples, the data plane mirror app tier **140** can facilitate direct communication between the control plane VCN **116** and the data plane VCN **118**. For example, changes, updates, or other suitable modifications to configuration may be desired to be applied to the resources contained in the data plane VCN **118**. Via a VNIC **142**, the control plane VCN **116** can directly communicate with, and can thereby execute the changes, updates, or other suitable modifications to configuration to, resources contained in the data plane VCN **118**.

[0055] In some embodiments, the control plane VCN **116** and the data plane VCN **118** can be contained in the service tenancy **119**. In this case, the user, or the customer, of the system may not own or operate either the control plane VCN **116** or the data plane VCN **118**. Instead, the IaaS provider may own or operate the control plane VCN **116** and the data plane VCN **118**, both of which may be contained in the service tenancy **119**. This embodiment can enable isolation of networks that may prevent users or customers from interacting with other users', or other customers', resources. Also, this embodiment may allow users or customers of the system to store databases privately without needing to rely on public Internet **154**, which may not have a desired level of threat prevention, for storage.

[0056] In other embodiments, the LB subnet(s) **122** contained in the control plane VCN **116** can be configured to receive a signal from the service gateway **136**. In this embodiment, the control plane VCN **116** and the data plane VCN **118** may be configured to be called by a customer of the IaaS provider without calling public Internet **154**. Customers of the IaaS provider may desire this embodiment since database(s) that the customers use may be controlled by the IaaS provider and may be stored on the service tenancy **119**, which may be isolated from public Internet **154**.

[0057] FIG. **2** is a block diagram **200** illustrating another example pattern of an IaaS architecture, according to at least one embodiment. Service operators **202** (e.g., service operators **102** of FIG. **1**) can be communicatively coupled to a secure host tenancy **204** (e.g., the secure host tenancy **104** of FIG. **1**) that can include a virtual cloud network (VCN) **206** (e.g., the VCN **106** of FIG. **1**) and a secure host subnet **208** (e.g., the secure host subnet **108** of FIG. **1**). The VCN **206** can include a local peering gateway (LPG) **210** (e.g., the LPG **110** of FIG. **1**) that can be communicatively coupled to a secure shell (SSH) VCN **212** (e.g., the SSH VCN **112** of FIG. **1**) via an LPG **110** contained in the SSH VCN **212**. The

SSH VCN **212** can include an SSH subnet **214** (e.g., the SSH subnet **114** of FIG. **1**), and the SSH VCN **212** can be communicatively coupled to a control plane VCN **216** (e.g., the control plane VCN **116** of FIG. **1**) via an LPG **210** contained in the control plane VCN **216**. The control plane VCN **216** can be contained in a service tenancy **219** (e.g., the service tenancy **119** of FIG. **1**), and the data plane VCN **218** (e.g., the data plane VCN **118** of FIG. **1**) can be contained in a customer tenancy **221** that may be owned or operated by users, or customers, of the system.

[0058] The control plane VCN **216** can include a control plane DMZ tier **220** (e.g., the control plane DMZ tier **120** of FIG. **1**) that can include LB subnet(s) **222** (e.g., LB subnet(s) **122** of FIG. **1**), a control plane app tier **224** (e.g., the control plane app tier **124** of FIG. **1**) that can include app subnet(s) **226** (e.g., app subnet(s) **126** of FIG. **1**), a control plane data tier **228** (e.g., the control plane data tier **128** of FIG. **1**) that can include database (DB) subnet(s) **230** (e.g., similar to DB subnet(s) **130** of FIG. **1**). The LB subnet(s) **222** contained in the control plane DMZ tier **220** can be communicatively coupled to the app subnet(s) **226** contained in the control plane app tier **224** and an Internet gateway **234** (e.g., the Internet gateway **134** of FIG. **1**) that can be contained in the control plane VCN **216**, and the app subnet(s) **226** can be communicatively coupled to the DB subnet(s) **230** contained in the control plane data tier **228** and a service gateway **236** (e.g., the service gateway **136** of FIG. **1**) and a network address translation (NAT) gateway **238** (e.g., the NAT gateway **138** of FIG. **1**). The control plane VCN **216** can include the service gateway **236** and the NAT gateway **238**.

[0059] The control plane VCN **216** can include a data plane mirror app tier **240** (e.g., the data plane mirror app tier **140** of FIG. **1**) that can include app subnet(s) **226**. The app subnet(s) **226** contained in the data plane mirror app tier **240** can include a virtual network interface controller (VNIC) **242** (e.g., the VNIC of **142**) that can execute a compute instance **244** (e.g., similar to the compute instance **144** of FIG. **1**). The compute instance **244** can facilitate communication between the app subnet(s) **226** of the data plane mirror app tier **240** and the app subnet(s) **226** that can be contained in a data plane app tier **246** (e.g., the data plane app tier **146** of FIG. **1**) via the VNIC **242** contained in the data plane mirror app tier **240** and the VNIC **242** contained in the data plane app tier **246**.

[0060] The Internet gateway **234** contained in the control plane VCN **216** can be communicatively coupled to a metadata management service **252** (e.g., the metadata management service **152** of FIG. **1**) that can be communicatively coupled to public Internet **254** (e.g., public Internet **154** of FIG. **1**). Public Internet **254** can be communicatively coupled to the NAT gateway **238** contained in the control plane VCN **216**. The service gateway **236** contained in the control plane VCN **216** can be communicatively coupled to cloud services **256** (e.g., cloud services **156** of FIG. **1**).

[0061] In some examples, the data plane VCN **218** can be contained in the customer tenancy **221**. In this case, the IaaS provider may provide the control plane VCN **216** for each customer, and the IaaS provider may, for each customer, set up a unique compute instance **244** that is contained in the service tenancy **219**. Each compute instance **244** may allow communication between the control plane VCN **216**, contained in the service tenancy **219**, and the data plane VCN **218** that is contained in the customer tenancy **221**. The compute instance **244** may allow resources, that are provisioned in the control plane VCN **216** that is contained in the service tenancy **219**, to be deployed or otherwise used in the data plane VCN **218** that is contained in the customer tenancy **221**.

[0062] In other examples, the customer of the IaaS provider may have databases that live in the customer tenancy **221**. In this example, the control plane VCN **216** can include the data plane mirror app tier **240** that can include app subnet(s) **226**. The data plane mirror app tier **240** can reside in the data plane VCN **218**, but the data plane mirror app tier **240** may not live in the data plane VCN **218**. That is, the data plane mirror app tier **240** may have access to the customer tenancy **221**, but the data plane mirror app tier **240** may not exist in the data plane VCN **218** or be owned or operated by the customer of the IaaS provider. The data plane mirror app tier **240** may be configured to make calls to the data plane VCN **218** but may not be configured to make calls to any entity contained in the control plane VCN **216**. The customer may desire to deploy or otherwise use resources in the data plane VCN **218** that are provisioned in the control plane VCN **216**, and the data plane mirror app tier **240** can facilitate the desired deployment, or other usage of resources, of the customer.

[0063] In some embodiments, the customer of the IaaS provider can apply filters to the data plane VCN **218**. In this embodiment, the customer can determine what the data plane VCN **218** can access, and the customer may restrict access to public Internet **254** from the data plane VCN **218**. The IaaS provider may not be able to apply filters or otherwise control access of the data plane VCN **218** to any outside networks or databases. Applying filters and controls by the customer onto the data plane VCN **218**, contained in the customer tenancy **221**, can help isolate the data plane VCN **218** from other customers and from public Internet **254**.

[0064] In some embodiments, cloud services **256** can be called by the service gateway **236** to access services that may not exist on public Internet **254**, on the control plane VCN **216**, or on the data plane VCN **218**. The connection between cloud services **256** and the control plane VCN **216** or the data plane VCN **218** may not be live or continuous. Cloud services **256** may exist on a different network owned or operated by the IaaS provider. Cloud services **256** may be configured to receive calls from the service gateway **236** and may be configured to not receive calls from public Internet **254**. Some cloud services **256** may be isolated from other cloud services **256**, and the control plane VCN **216** may be isolated from cloud services **256** that may not be in the same region as the control plane VCN **216**. For example, the control plane VCN **216** may be located in "Region 1," and cloud service "Deployment 1," may be located in Region 1 and in "Region 2." If a call to Deployment 1 is made by the service gateway **236** contained in the control plane VCN **216** located in Region 1, the call may be transmitted to Deployment 1 in Region 1. In this example, the control plane VCN **216**, or Deployment 1 in Region 1, may not be communicatively coupled to, or otherwise in communication with, Deployment 1 in Region 2.

[0065] FIG. **3** is a block diagram **300** illustrating another example pattern of an IaaS architecture, according to at least one embodiment. Service operators **302** (e.g., service operators **102** of FIG. **1**) can be communicatively coupled to a secure host tenancy **304** (e.g., the secure host tenancy **104** of FIG. **1**) that can include a virtual cloud network (VCN) **306**

(e.g., the VCN **106** of FIG. **1**) and a secure host subnet **308** (e.g., the secure host subnet **108** of FIG. **1**). The VCN **306** can include an LPG **310** (e.g., the LPG **110** of FIG. **1**) that can be communicatively coupled to an SSH VCN **312** (e.g., the SSH VCN **112** of FIG. **1**) via an LPG **310** contained in the SSH VCN **312**. The SSH VCN **312** can include an SSH subnet **314** (e.g., the SSH subnet **114** of FIG. **1**), and the SSH VCN **312** can be communicatively coupled to a control plane VCN **316** (e.g., the control plane VCN **116** of FIG. **1**) via an LPG **310** contained in the control plane VCN **316** and to a data plane VCN **318** (e.g., the data plane **118** of FIG. **1**) via an LPG **310** contained in the data plane VCN **318**. The control plane VCN **316** and the data plane VCN **318** can be contained in a service tenancy **319** (e.g., the service tenancy **119** of FIG. **1**).

[0066] The control plane VCN **316** can include a control plane DMZ tier **320** (e.g., the control plane DMZ tier **120** of FIG. **1**) that can include load balancer (LB) subnet(s) **322** (e.g., LB subnet(s) **122** of FIG. **1**), a control plane app tier **324** (e.g., the control plane app tier **124** of FIG. **1**) that can include app subnet(s) **326** (e.g., similar to app subnet(s) **126** of FIG. **1**), a control plane data tier **328** (e.g., the control plane data tier **128** of FIG. **1**) that can include DB subnet(s) **330**. The LB subnet(s) **322** contained in the control plane DMZ tier **320** can be communicatively coupled to the app subnet(s) **326** contained in the control plane app tier **324** and to an Internet gateway **334** (e.g., the Internet gateway **134** of FIG. **1**) that can be contained in the control plane VCN **316**, and the app subnet(s) **326** can be communicatively coupled to the DB subnet(s) **330** contained in the control plane data tier **328** and to a service gateway **336** (e.g., the service gateway of FIG. **1**) and a network address translation (NAT) gateway **338** (e.g., the NAT gateway **138** of FIG. **1**). The control plane VCN **316** can include the service gateway **336** and the NAT gateway **338**.

[0067] The data plane VCN **318** can include a data plane app tier **346** (e.g., the data plane app tier **146** of FIG. **1**), a data plane DMZ tier **348** (e.g., the data plane DMZ tier **148** of FIG. **1**), and a data plane data tier **350** (e.g., the data plane data tier **150** of FIG. **1**). The data plane DMZ tier **348** can include LB subnet(s) **322** that can be communicatively coupled to trusted app subnet(s) **360** and untrusted app subnet(s) **362** of the data plane app tier **346** and the Internet gateway **334** contained in the data plane VCN **318**. The trusted app subnet(s) **360** can be communicatively coupled to the service gateway **336** contained in the data plane VCN **318**, the NAT gateway **338** contained in the data plane VCN **318**, and DB subnet(s) **330** contained in the data plane data tier **350**. The untrusted app subnet(s) **362** can be communicatively coupled to the service gateway **336** contained in the data plane VCN **318** and DB subnet(s) **330** contained in the data plane data tier **350**. The data plane data tier **350** can include DB subnet(s) **330** that can be communicatively coupled to the service gateway **336** contained in the data plane VCN **318**.

[0068] The untrusted app subnet(s) **362** can include one or more primary VNICs **364**(**1**)-(N) that can be communicatively coupled to tenant virtual machines (VMs) **366**(**1**)-(N). Each tenant VM **366**(**1**)-(N) can be communicatively coupled to a respective app subnet **367**(**1**)-(N) that can be contained in respective container egress VCNs **368**(**1**)-(N) that can be contained in respective customer tenancies **370**(**1**)-(N). Respective secondary VNICs **372**(**1**)-(N) can facilitate communication between the untrusted app subnet(s) **362** contained in the data plane VCN **318** and the app subnet contained in the container egress VCNs **368**(**1**)-(N). Each container egress VCNs **368**(**1**)-(N) can include a NAT gateway **338** that can be communicatively coupled to public Internet **354** (e.g., public Internet **154** of FIG. **1**).

[0069] The Internet gateway **334** contained in the control plane VCN **316** and contained in the data plane VCN **318** can be communicatively coupled to a metadata management service **352** (e.g., the metadata management system **152** of FIG. **1**) that can be communicatively coupled to public Internet **354**. Public Internet **354** can be communicatively coupled to the NAT gateway **338** contained in the control plane VCN **316** and contained in the data plane VCN **318**. The service gateway **336** contained in the control plane VCN **316** and contained in the data plane VCN **318** can be communicatively coupled to cloud services **356**.

[0070] In some embodiments, the data plane VCN **318** can be integrated with customer tenancies **370**. This integration can be useful or desirable for customers of the IaaS provider in some cases such as a case that may desire support when executing code. The customer may provide code to run that may be destructive, may communicate with other customer resources, or may otherwise cause undesirable effects. In response to this, the IaaS provider may determine whether to run code given to the IaaS provider by the customer.

[0071] In some examples, the customer of the IaaS provider may grant temporary network access to the IaaS provider and request a function to be attached to the data plane app tier **346**. Code to run the function may be executed in the VMs **366**(**1**)-(N), and the code may not be configured to run anywhere else on the data plane VCN **318**. Each VM **366**(**1**)-(N) may be connected to one customer tenancy **370**. Respective containers **371**(**1**)-(N) contained in the VMs **366**(**1**)-(N) may be configured to run the code. In this case, there can be a dual isolation (e.g., the containers **371**(**1**)-(N) running code, where the containers **371**(**1**)-(N) may be contained in at least the VM **366**(**1**)-(N) that are contained in the untrusted app subnet(s) **362**), which may help prevent incorrect or otherwise undesirable code from damaging the network of the IaaS provider or from damaging a network of a different customer. The containers **371**(**1**)-(N) may be communicatively coupled to the customer tenancy **370** and may be configured to transmit or receive data from the customer tenancy **370**. The containers **371**(**1**)-(N) may not be configured to transmit or receive data from any other entity in the data plane VCN **318**. Upon completion of running the code, the IaaS provider may kill or otherwise dispose of the containers **371**(**1**)-(N).

[0072] In some embodiments, the trusted app subnet(s) **360** may run code that may be owned or operated by the IaaS provider. In this embodiment, the trusted app subnet(s) **360** may be communicatively coupled to the DB subnet(s) **330** and be configured to execute CRUD operations in the DB subnet(s) **330**. The untrusted app subnet(s) **362** may be communicatively coupled to the DB subnet(s) **330**, but in this embodiment, the untrusted app subnet(s) may be configured to execute read operations in the DB subnet(s) **330**. The containers **371**(**1**)-(N) that can be contained in the VM **366**(**1**)-(N) of each customer and that may run code from the customer may not be communicatively coupled with the DB subnet(s) **330**.

[0073] In other embodiments, the control plane VCN **316** and the data plane VCN **318** may not be directly communicatively coupled. In this embodiment, there may be no

direct communication between the control plane VCN **316** and the data plane VCN **318**. However, communication can occur indirectly through at least one method. An LPG **310** may be established by the IaaS provider that can facilitate communication between the control plane VCN **316** and the data plane VCN **318**. In another example, the control plane VCN **316** or the data plane VCN **318** can make a call to cloud services **356** via the service gateway **336**. For example, a call to cloud services **356** from the control plane VCN **316** can include a request for a service that can communicate with the data plane VCN **318**.

[0074] FIG. **4** is a block diagram **400** illustrating another example pattern of an IaaS architecture, according to at least one embodiment. Service operators **402** (e.g., service operators **102** of FIG. **1**) can be communicatively coupled to a secure host tenancy **404** (e.g., the secure host tenancy **104** of FIG. **1**) that can include a virtual cloud network (VCN) **406** (e.g., the VCN **106** of FIG. **1**) and a secure host subnet **408** (e.g., the secure host subnet **108** of FIG. **1**). The VCN **406** can include an LPG **410** (e.g., the LPG **110** of FIG. **1**) that can be communicatively coupled to an SSH VCN **412** (e.g., the SSH VCN **112** of FIG. **1**) via an LPG **410** contained in the SSH VCN **412**. The SSH VCN **412** can include an SSH subnet **414** (e.g., the SSH subnet **114** of FIG. **1**), and the SSH VCN **412** can be communicatively coupled to a control plane VCN **416** (e.g., the control plane VCN **116** of FIG. **1**) via an LPG **410** contained in the control plane VCN **416** and to a data plane VCN **418** (e.g., the data plane **118** of FIG. **1**) via an LPG **410** contained in the data plane VCN **418**. The control plane VCN **416** and the data plane VCN **418** can be contained in a service tenancy **419** (e.g., the service tenancy **119** of FIG. **1**).

[0075] The control plane VCN **416** can include a control plane DMZ tier **420** (e.g., the control plane DMZ tier **120** of FIG. **1**) that can include LB subnet(s) **422** (e.g., LB subnet(s) **122** of FIG. **1**), a control plane app tier **424** (e.g., the control plane app tier **124** of FIG. **1**) that can include app subnet(s) **426** (e.g., app subnet(s) **126** of FIG. **1**), a control plane data tier **428** (e.g., the control plane data tier **128** of FIG. **1**) that can include DB subnet(s) **430** (e.g., DB subnet(s) **330** of FIG. **3**). The LB subnet(s) **422** contained in the control plane DMZ tier **420** can be communicatively coupled to the app subnet(s) **426** contained in the control plane app tier **424** and to an Internet gateway **434** (e.g., the Internet gateway **134** of FIG. **1**) that can be contained in the control plane VCN **416**, and the app subnet(s) **426** can be communicatively coupled to the DB subnet(s) **430** contained in the control plane data tier **428** and to a service gateway **436** (e.g., the service gateway of FIG. **1**) and a network address translation (NAT) gateway **438** (e.g., the NAT gateway **138** of FIG. **1**). The control plane VCN **416** can include the service gateway **436** and the NAT gateway **438**.

[0076] The data plane VCN **418** can include a data plane app tier **446** (e.g., the data plane app tier **146** of FIG. **1**), a data plane DMZ tier **448** (e.g., the data plane DMZ tier **148** of FIG. **1**), and a data plane data tier **450** (e.g., the data plane data tier **150** of FIG. **1**). The data plane DMZ tier **448** can include LB subnet(s) **422** that can be communicatively coupled to trusted app subnet(s) **460** (e.g., trusted app subnet(s) **360** of FIG. **3**) and untrusted app subnet(s) **462** (e.g., untrusted app subnet(s) **362** of FIG. **3**) of the data plane app tier **446** and the Internet gateway **434** contained in the data plane VCN **418**. The trusted app subnet(s) **460** can be communicatively coupled to the service gateway **436** contained in the data plane VCN **418**, the NAT gateway **438** contained in the data plane VCN **418**, and DB subnet(s) **430** contained in the data plane data tier **450**. The untrusted app subnet(s) **462** can be communicatively coupled to the service gateway **436** contained in the data plane VCN **418** and DB subnet(s) **430** contained in the data plane data tier **450**. The data plane data tier **450** can include DB subnet(s) **430** that can be communicatively coupled to the service gateway **436** contained in the data plane VCN **418**.

[0077] The untrusted app subnet(s) **462** can include primary VNICs **464**(**1**)-(N) that can be communicatively coupled to tenant virtual machines (VMs) **466**(**1**)-(N) residing within the untrusted app subnet(s) **462**. Each tenant VM **466**(**1**)-(N) can run code in a respective container **467**(**1**)-(N), and be communicatively coupled to an app subnet **426** that can be contained in a data plane app tier **446** that can be contained in a container egress VCN **468**. Respective secondary VNICs **472**(**1**)-(N) can facilitate communication between the untrusted app subnet(s) **462** contained in the data plane VCN **418** and the app subnet contained in the container egress VCN **468**. The container egress VCN can include a NAT gateway **438** that can be communicatively coupled to public Internet **454** (e.g., public Internet **154** of FIG. **1**).

[0078] The Internet gateway **434** contained in the control plane VCN **416** and contained in the data plane VCN **418** can be communicatively coupled to a metadata management service **452** (e.g., the metadata management system **152** of FIG. **1**) that can be communicatively coupled to public Internet **454**. Public Internet **454** can be communicatively coupled to the NAT gateway **438** contained in the control plane VCN **416** and contained in the data plane VCN **418**. The service gateway **436** contained in the control plane VCN **416** and contained in the data plane VCN **418** can be communicatively coupled to cloud services **456**.

[0079] In some examples, the pattern illustrated by the architecture of block diagram **400** of FIG. **4** may be considered an exception to the pattern illustrated by the architecture of block diagram **300** of FIG. **3** and may be desirable for a customer of the IaaS provider if the IaaS provider cannot directly communicate with the customer (e.g., a disconnected region). The respective containers **467**(**1**)-(N) that are contained in the VMs **466**(**1**)-(N) for each customer can be accessed in real-time by the customer. The containers **467**(**1**)-(N) may be configured to make calls to respective secondary VNICs **472**(**1**)-(N) contained in app subnet(s) **426** of the data plane app tier **446** that can be contained in the container egress VCN **468**. The secondary VNICs **472**(**1**)-(N) can transmit the calls to the NAT gateway **438** that may transmit the calls to public Internet **454**. In this example, the containers **467**(**1**)-(N) that can be accessed in real-time by the customer can be isolated from the control plane VCN **416** and can be isolated from other entities contained in the data plane VCN **418**. The containers **467**(**1**)-(N) may also be isolated from resources from other customers.

[0080] In other examples, the customer can use the containers **467**(**1**)-(N) to call cloud services **456**. In this example, the customer may run code in the containers **467**(**1**)-(N) that requests a service from cloud services **456**. The containers **467**(**1**)-(N) can transmit this request to the secondary VNICs **472**(**1**)-(N) that can transmit the request to the NAT gateway that can transmit the request to public Internet **454**. Public Internet **454** can transmit the request to LB subnet(s) **422** contained in the control plane VCN **416**

via the Internet gateway **434**. In response to determining the request is valid, the LB subnet(s) can transmit the request to app subnet(s) **426** that can transmit the request to cloud services **456** via the service gateway **436**.

[0081] It should be appreciated that IaaS architectures **100**, **200**, **300**, **400** depicted in the figures may have other components than those depicted. Further, the embodiments shown in the figures are only some examples of a cloud infrastructure system that may incorporate an embodiment of the disclosure. In some other embodiments, the IaaS systems may have more or fewer components than shown in the figures, may combine two or more components, or may have a different configuration or arrangement of components.

[0082] In certain embodiments, the IaaS systems described herein may include a suite of applications, middleware, and database service offerings that are delivered to a customer in a self-service, subscription-based, elastically scalable, reliable, highly available, and secure manner. An example of such an IaaS system is the Oracle Cloud Infrastructure (OCI) provided by the present assignee.

Computer System

[0083] FIG. **5** illustrates an example computer system **500**, in which various embodiments described herein may be implemented. The system **500** may be used to implement any of the computer systems described above. As shown in the figure, computer system **500** includes a processing unit **504** that communicates with a number of peripheral subsystems via a bus subsystem **502**. These peripheral subsystems may include a processing acceleration unit **506**, an I/O subsystem **508**, a storage subsystem **518** and a communications subsystem **524**. Storage subsystem **518** includes tangible computer-readable storage media **522** and a system memory **510**.

[0084] Bus subsystem **502** provides a mechanism for letting the various components and subsystems of computer system **500** communicate with each other as intended. Although bus subsystem **502** is shown schematically as a single bus, alternative embodiments of the bus subsystem may utilize multiple buses. Bus subsystem **502** may be any of several types of bus structures including a memory bus or memory controller, a peripheral bus, and a local bus using any of a variety of bus architectures. For example, such architectures may include an Industry Standard Architecture (ISA) bus, Micro Channel Architecture (MCA) bus, Enhanced ISA (EISA) bus, Video Electronics Standards Association (VESA) local bus, and Peripheral Component Interconnect (PCI) bus, which can be implemented as a Mezzanine bus manufactured to the IEEE P1386.1 standard.

[0085] Processing unit **504**, which can be implemented as one or more integrated circuits (e.g., a conventional microprocessor or microcontroller), controls the operation of computer system **500**. One or more processors may be included in processing unit **504**. These processors may include single core or multicore processors. In certain embodiments, processing unit **504** may be implemented as one or more independent processing units **532** and/or **534** with single or multicore processors included in each processing unit. In other embodiments, processing unit **504** may also be implemented as a quad-core processing unit formed by integrating two dual-core processors into a single chip.

[0086] In various embodiments, processing unit **504** can execute a variety of programs in response to program code and can maintain multiple concurrently executing programs or processes. At any given time, some or all of the program code to be executed can be resident in processor(s) **504** and/or in storage subsystem **518**. Through suitable programming, processor(s) **504** can provide various functionalities described above. Computer system **500** may additionally include a processing acceleration unit **506**, which can include a digital signal processor (DSP), a special-purpose processor, and/or the like.

[0087] I/O subsystem **508** may include user interface input devices and user interface output devices. User interface input devices may include a keyboard, pointing devices such as a mouse or trackball, a touchpad or touch screen incorporated into a display, a scroll wheel, a click wheel, a dial, a button, a switch, a keypad, audio input devices with voice command recognition systems, microphones, and other types of input devices. User interface input devices may include, for example, motion sensing and/or gesture recognition devices such as the Microsoft Kinect® motion sensor that enables users to control and interact with an input device, such as the Microsoft Xbox® 360 game controller, through a natural user interface using gestures and spoken commands. User interface input devices may also include eye gesture recognition devices such as the Google Glass® blink detector that detects eye activity (e.g., 'blinking' while taking pictures and/or making a menu selection) from users and transforms the eye gestures as input into an input device (e.g., Google Glass®). Additionally, user interface input devices may include voice recognition sensing devices that enable users to interact with voice recognition systems (e.g., Siri® navigator), through voice commands.

[0088] User interface input devices may also include, without limitation, three dimensional (3D) mice, joysticks or pointing sticks, gamepads and graphic tablets, and audio/visual devices such as speakers, digital cameras, digital camcorders, portable media players, webcams, image scanners, fingerprint scanners, barcode reader 3D scanners, 3D printers, laser rangefinders, and eye gaze tracking devices. Additionally, user interface input devices may include, for example, medical imaging input devices such as computed tomography, magnetic resonance imaging, position emission tomography, medical ultrasonography devices. User interface input devices may also include, for example, audio input devices such as MIDI keyboards, digital musical instruments and the like.

[0089] User interface output devices may include a display subsystem, indicator lights, or non-visual displays such as audio output devices, etc. The display subsystem may be a cathode ray tube (CRT), a flat-panel device, such as that using a liquid crystal display (LCD) or plasma display, a projection device, a touch screen, and the like. In general, use of the term "output device" is intended to include all possible types of devices and mechanisms for outputting information from computer system **500** to a user or other computer. For example, user interface output devices may include, without limitation, a variety of display devices that visually convey text, graphics and audio/video information such as monitors, printers, speakers, headphones, automotive navigation systems, plotters, voice output devices, and modems.

[0090] Computer system **500** may comprise a storage subsystem **518** that provides a tangible non-transitory computer-readable storage medium for storing software and data constructs that provide the functionality of the embodiments

described in this disclosure. The software can include programs, code modules, instructions, scripts, etc., that when executed by one or more cores or processors of processing unit **504** provide the functionality described above. Storage subsystem **518** may also provide a repository for storing data used in accordance with the present disclosure.

[0091] As depicted in the example in FIG. **5**, storage subsystem **518** can include various components including a system memory **510**, computer-readable storage media **522**, and a computer readable storage media reader **520**. System memory **510** may store program instructions that are loadable and executable by processing unit **504**. System memory **510** may also store data that is used during the execution of the instructions and/or data that is generated during the execution of the program instructions. Various different kinds of programs may be loaded into system memory **510** including but not limited to client applications, Web browsers, mid-tier applications, relational database management systems (RDBMS), virtual machines, containers, etc.

[0092] System memory **510** may also store an operating system **516**. Examples of operating system **516** may include various versions of Microsoft Windows®, Apple Macintosh®, and/or Linux operating systems, a variety of commercially-available UNIX® or UNIX-like operating systems (including without limitation the variety of GNU/Linux operating systems, the Google Chrome® OS, and the like) and/or mobile operating systems such as iOS, Windows® Phone, Android® OS, BlackBerry® OS, and Palm® OS operating systems. In certain implementations where computer system **500** executes one or more virtual machines, the virtual machines along with their guest operating systems (GOSs) may be loaded into system memory **510** and executed by one or more processors or cores of processing unit **504**.

[0093] System memory **510** can come in different configurations depending upon the type of computer system **500**. For example, system memory **510** may be volatile memory (such as random access memory (RAM)) and/or non-volatile memory (such as read-only memory (ROM), flash memory, etc.) Different types of RAM configurations may be provided including a static random access memory (SRAM), a dynamic random access memory (DRAM), and others. In some implementations, system memory **510** may include a basic input/output system (BIOS) containing basic routines that help to transfer information between elements within computer system **500**, such as during start-up.

[0094] Computer-readable storage media **522** may represent remote, local, fixed, and/or removable storage devices plus storage media for temporarily and/or more permanently containing, storing, computer-readable information for use by computer system **500** including instructions executable by processing unit **504** of computer system **500**.

[0095] Computer-readable storage media **522** can include any appropriate media known or used in the art, including storage media and communication media, such as but not limited to, volatile and non-volatile, removable and non-removable media implemented in any method or technology for storage and/or transmission of information. This can include tangible computer-readable storage media such as RAM, ROM, electronically erasable programmable ROM (EEPROM), flash memory or other memory technology, CD-ROM, digital versatile disk (DVD), or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or other tangible computer readable media.

[0096] By way of example, computer-readable storage media **522** may include a hard disk drive that reads from or writes to non-removable, nonvolatile magnetic media, a magnetic disk drive that reads from or writes to a removable, nonvolatile magnetic disk, and an optical disk drive that reads from or writes to a removable, nonvolatile optical disk such as a CD ROM, DVD, and Blu-Ray® disk, or other optical media. Computer-readable storage media **522** may include, but is not limited to, Zip® drives, flash memory cards, universal serial bus (USB) flash drives, secure digital (SD) cards, DVD disks, digital video tape, and the like. Computer-readable storage media **522** may also include, solid-state drives (SSD) based on non-volatile memory such as flash-memory based SSDs, enterprise flash drives, solid state ROM, and the like, SSDs based on volatile memory such as solid state RAM, dynamic RAM, static RAM, DRAM-based SSDs, magnetoresistive RAM (MRAM) SSDs, and hybrid SSDs that use a combination of DRAM and flash memory based SSDs. The disk drives and their associated computer-readable media may provide non-volatile storage of computer-readable instructions, data structures, program modules, and other data for computer system **500**.

[0097] Machine-readable instructions executable by one or more processors or cores of processing unit **504** may be stored on a non-transitory computer-readable storage medium. A non-transitory computer-readable storage medium can include physically tangible memory or storage devices that include volatile memory storage devices and/or non-volatile storage devices. Examples of non-transitory computer-readable storage medium include magnetic storage media (e.g., disk or tapes), optical storage media (e.g., DVDs, CDs), various types of RAM, ROM, or flash memory, hard drives, floppy drives, detachable memory drives (e.g., USB drives), or other type of storage device.

[0098] Communications subsystem **524** provides an interface to other computer systems and networks. Communications subsystem **524** serves as an interface for receiving data from and transmitting data to other systems from computer system **500**. For example, communications subsystem **524** may enable computer system **500** to connect to one or more devices via the Internet. In some embodiments communications subsystem **524** can include radio frequency (RF) transceiver components for accessing wireless voice and/or data networks (e.g., using cellular telephone technology, advanced data network technology, such as 3G, 4G or EDGE (enhanced data rates for global evolution), WiFi (IEEE 802.11 family standards, or other mobile communication technologies, or any combination thereof), global positioning system (GPS) receiver components, and/or other components. In some embodiments communications subsystem **524** can provide wired network connectivity (e.g., Ethernet) in addition to or instead of a wireless interface.

[0099] In some embodiments, communications subsystem **524** may also receive input communication in the form of structured and/or unstructured data feeds **526**, event streams **528**, event updates **530**, and the like on behalf of one or more users who may use computer system **500**.

[0100] By way of example, communications subsystem **524** may be configured to receive data feeds **526** in real-time from users of social networks and/or other communication

services such as Twitter® feeds, Facebook® updates, web feeds such as Rich Site Summary (RSS) feeds, and/or real-time updates from one or more third party information sources.

[0101] Additionally, communications subsystem **524** may also be configured to receive data in the form of continuous data streams, which may include event streams **528** of real-time events and/or event updates **530**, that may be continuous or unbounded in nature with no explicit end. Examples of applications that generate continuous data may include, for example, sensor data applications, financial tickers, network performance measuring tools (e.g., network monitoring and traffic management applications), click-stream analysis tools, automobile traffic monitoring, and the like.

[0102] Communications subsystem **524** may also be configured to output the structured and/or unstructured data feeds **526**, event streams **528**, event updates **530**, and the like to one or more databases that may be in communication with one or more streaming data source computers coupled to computer system **500**.

[0103] Computer system **500** can be one of various types, including a handheld portable device (e.g., an iPhone® cellular phone, an iPad® computing tablet, a PDA), a wearable device (e.g., a Google Glass® head mounted display), a PC, a workstation, a mainframe, a kiosk, a server rack, or any other data processing system.

[0104] Due to the ever-changing nature of computers and networks, the description of computer system **500** depicted in the figure is intended only as a specific example. Many other configurations having more or fewer components than the system depicted in the figure are possible. For example, customized hardware might also be used and/or particular elements might be implemented in hardware, firmware, software (including applets), or a combination. Further, connection to other computing devices, such as network input/output devices, may be employed. Based on the disclosure and teachings provided herein, a person of ordinary skill in the art will appreciate other ways and/or methods to implement the various embodiments.

AI Training Using Accessibility Data

[0105] As described above, an end-user (also referred to herein as a "user") may rely on a helpdesk service (e.g., provided within a cloud computing environment) to troubleshoot an error state occurring on the end-user device (also referred to herein as an "endpoint device"). FIG. **6** is a block diagram illustrating a system architecture **600** for training an AI model to troubleshoot error states on user devices, such as the at least one user device **604** illustrated in FIG. **6**, according to at least one embodiment. In the illustrated example, the at least one user device **604** includes a first user device **604***a* and a second user device **604***b*. However, the system architecture **600** may include more than two user devices **604** or less than two user devices **604**.

[0106] Each user device **604** may be substantially similar to the computer system **500** described above with respect to FIG. **5**. For example, each user device **604** may include a processing unit substantially similar to the processing unit **504**, a processing acceleration unit substantially similar to the processing acceleration unit **506**, a I/O subsystem substantially similar to the I/O subsystem **508**, a storage subsystem substantially similar to the storage subsystem **518**, and/or a communications subsystem substantially similar to the communications subsystem **524**.

[0107] Each user device **604**, or endpoint device **604**, is provided with an endpoint agent **608** configured to receive accessibility data from an accessibility data source **612**. The endpoint agent **608** receives the accessibility data by making one or more calls via an accessibility application programming interface (API), otherwise referred to herein as an accessibility interface, to request the accessibility data. Calls made via the accessibility API are calls that conform to the accessibility API, which may conform to an accessibility standard.

[0108] The accessibility data source **612** is a platform that renders a user interface ("UI") displayed by the user device **604**. For example, the accessibility data source **612** may include an operating system ("OS") executing on the user device **604**, a browser application executing on the user device **604**, and/or another software application executing on the user device **604**. Accordingly, the accessibility data is generated by the platform executing on the user device **604** during rendering of the UI.

[0109] The accessibility data, which is associated with the UI displayed by the user device **604**, includes information identifying UI elements in the UI of the user device **604** and/or information identifying UI events in the UI of the user device **604**. UI elements may include, for example, windows, buttons, scroll bars, menus, and/or the like. UI events include changes in a state or condition of the UI elements. UI events may be caused by keyboard strokes, mouse clicks, software execution, and/or the like. For example, keyboard strokes may cause text rendered in a UI element to change.

[0110] The accessibility data may also be received by an assistive application to perform assistive, or accessibility, functionality. The accessibility functionality may be, for example, functionality beyond that which is offered by the UI rendered on the user device **604** to facilitate user interaction with the UI (e.g., by users with disabilities). Assistive applications may include, but are not limited to, screen magnifiers, screen readers, text-to-speech software, speech recognition software, and/or the like.

[0111] The accessibility data (i.e., the information identifying UI elements and/or the information identifying UI events in the UI of the user device **604**) is generated based on metadata exposed via the accessibility API. Metadata exposed via the accessibility API is metadata that conforms to the accessibility API, which may conform to a standard, such as an accessibility or UI standard. Metadata generally refers to labels created by, for example, a developer to identify characteristics of UI elements.

[0112] In some instances, the information identifying the UI elements in a UI of the user device **604** is organized as a hierarchical tree. For example, FIGS. **7**A and **7**B illustrate example accessibility data. FIG. **7**A illustrates an example accessibility data tree **700** of accessibility data associated with UI elements of an example UI of user device **604** (i.e., any one of the first user device **604***a* or the second user device **604***b*). Each hierarchical level of the hierarchical tree **700** includes at least one node respectively representing a UI element of the example UI and associated metadata for each element. The hierarchical relationships between respective UI elements of the example UI may be specified by the metadata.

[0113] For each respective UI element, the metadata may also specify a respective element type of the respective UI element and/or a respective identifier of the respective UI element. For example, as indicated by the hierarchical tree **700**, a UI element of the example UI includes a pane-type element **704** identified as "Desktop."

[0114] Each respective UI element may include one or more secondary UI elements. For example, the pane-type UI element may include a first window-type UI element **708** identified as "Window1" and a second window-type UI element **710** identified as "Window2" that are each at a hierarchical level of the hierarchical tree **700** below the pane-type UI element **704**. In other words, each hierarchical level of the hierarchical tree **700** includes at least one node respectively representing a UI element.

[0115] As further illustrated in FIG. **7**A, the first window-type UI element **708** includes, at a lower hierarchical level than the pane-type UI element **708**, a TitleBar-type UI element **712** identified as "TitleBar1," and defining a title bar portion of the window-type UI element **708**, a menu-type UI element **716** identified as "Menu1" and defining a menu of the window-type UI element **708**, a StatusBar-type UI element **720** identified as "StatusBar1" and defining a status bar of the window-type UI element **708**, and a group-type UI element **724** identified as "Group1," and defining a UI group included in the window-type UI element **708**. The group-type UI element **724** further includes a text-type UI element **728** identified as "Group1," defining text displayed within the group-type UI element **724**, and a tree-type UI element **732** displayed within the group-type UI element **724**.

[0116] For each respective UI element, the metadata may also specify a respective state or condition of the UI element. For example, as illustrated in FIG. **7**B, the accessibility data (e.g., the metadata) may also provide additional metadata of a particular UI element (e.g., the tree-type UI element **732** of FIG. **7**A). As illustrated in FIG. **7**B, additional metadata **740** defining states or conditions of a given UI element may include other identifiers (e.g., an automation identifier, a name, a class name), control information (e.g., a control type, a localized control type), framework information (e.g., a framework type, a framework identifier), process information (e.g., a process identifier), location information of the UI element within the example UI (e.g., coordinates and/or dimensions of the UI element), pattern information of the UI element, and/or the like.

[0117] Conditions and/or states of a respective UI element may change, for example, responsive to user interaction with the UI element, responsive to software execution, and/or the like. For example, responsive to a user interacting with a respective UI element to resize the UI element, bounding box information associated with the UI element may change (e.g., coordinates and/or dimensions of the UI element). As described above, UI events include changes in a state or condition of the UI elements. The information identifying UI events (i.e., information included as accessibility data requested by, for example, the endpoint agent **608**) may include notifications of such changes in states or conditions of the UI elements.

[0118] The metadata associated with the UI elements and/or UI events may be specified in application code of application for which the example UI is being rendered and/or content code of web content for which the example UI is being rendered.

[0119] Referring again to FIG. **6**, the endpoint agent **608** of each user device **604** is also configured to receive endpoint management data from an endpoint management data source **616** of the user device **604**. The endpoint agent **608** may request the endpoint management data by making one or more API calls to the endpoint management data source **616** via an endpoint management API.

[0120] The endpoint management data includes state information of the user device **604**. For example, the endpoint management data may include configuration information of the user device **604** (how the user device workstation is configured), network information of the user device **604** (e.g., whether and how a virtual private network ("VPN") is established, Wi-Fi connection information of the user device **604**, Ethernet connection information of the user device **604**, etc.), user profile information of a logged-in session of the user device **604**, process information of the user device **604** (e.g., programs run on and files accessed by the user device **604**), and/or other telemetry and log information associated with the user device **604**.

[0121] Each endpoint agent **608** may provide the requested accessibility data received from the accessibility data source **612** and the endpoint management data received from the endpoint management data source **616** to a server agent **620** residing in a server **624**. The server **624** may be an on-site server **624** within the same local network as each user device **604**, a cloud-based server **624**, or a combination thereof (i.e., a server **624** implemented in a distributed manner). In some instances, the server **624** is implemented as a computer system substantially similar to the computer system **500** described above with respect to FIG. **5**. Further, in some instances, functionality provided by the server **624** is included as part of the cloud services **456** described above with respect to FIG. **4**.

[0122] In some instances, the server agent **620** periodically receives the endpoint management data and/or the accessibility data from each user device **604** (e.g., at predetermined intervals of time). In some instances, the server agent **620** receives the endpoint management data and/or accessibility data from each user device **604** responsive to transmitting a command to the endpoint agent **608** requesting the endpoint management data and/or accessibility data.

[0123] For simplicity, functions are described herein as being performed by the server agent **620** or the endpoint agent **608**. However, such functions may be performed interchangeably by the endpoint agent **608** or the server agent **620**. For example, in some instances, the server agent **620** itself makes the respective API calls to the accessibility data source **612** and/or the endpoint management data source **616**. In some instances, rather than the server agent **620** requesting accessibility data from the accessibility data source **612** (e.g., via respective API calls), the accessibility data source **612** pushes the accessibility data to the server agent **620** (e.g., via one or more API calls).

[0124] As described above, a user may experience an error in the functionality of an end-user device, such as the first user device **604***a* of FIG. **6**. The error may be, for example, an error state of software running on the first user device **604***a*. The server agent **620** provides the error state information and the accessibility data of the first user device **604***a* as training data to an AI model **628** residing in the server **624**. The AI model may be, for example, a large language model ("LLM") that is fine tuned and/or prompt-tuned according to a curated set of error states (e.g., error states

extrapolated from accessibility data and/or helpdesk error reporting data), context surrounding those example error states (e.g., endpoint management data and/or accessibility data), and error state solutions. The LLM may be a suitably large model having a minimum number of parameters (e.g., at least 7 billion parameters, at least 25 billion parameters, at least 50 billion parameters, etc.). To fine-tune a pre-trained model, a training data set including the information noted above can be provided to the model (e.g., by defining a path to such training data, which may be stored in one or more object storage buckets on a cloud platform).

[0125] Based on the received accessibility data, the AI model **628** may identify error information displayed on the UI of the first user device **604***a* (e.g., an error window UI element including an error description, an error code, etc. as indicated by metadata of the UI element) and relate the displayed error information to the determined error state. Accordingly, the AI model **628** is trained to infer later occurrences of the error state on both the first user device **604***a* and other user devices **604** (e.g., the second user device **604***b*) based on later-received accessibility data.

[0126] For example, using the AI model **628**, the server agent **620** may determine that second accessibility data received from the second user device **604***b* (e.g., by way of the server agent **620** and the endpoint agent **608***b*) includes a similar error window UI having the same error code as previously received by the first user device **604***a* when the first user device **604***a* experienced a first error state. Accordingly, based on inferences of the AI model **628**, the server agent **620** may detect that the second user device **604***b* is experiencing an error state that is the same as the first error state previously experienced by the first user device **604***a*.

[0127] In some instances, the error state information provided by the server agent **620** to the AI model **628** as training data is received through a virtual helpdesk service, such as the helpdesk **632** illustrated in FIG. **6**. For example, responsive to the first user device **604***a* experiencing an error, a user may establish a connection to the virtual helpdesk **632** with the first user device **604***a* in order to report and troubleshoot the error state. The virtual helpdesk **632** may be an internal helpdesk service associated with an organization to which the first user device **604***a* and the second user device **604***b* belong. The server agent **620** may be communicatively connected to the virtual helpdesk **632** such that the server agent **620** receives communications between the virtual helpdesk **632** and the first user device **604***a*. For example, the server agent **620** may monitor the helpdesk **632**, or receive a notification from the helpdesk **632** responsive to an error being reported to the helpdesk **632**.

[0128] In some instances, the AI model **628** is further trained based on a solution to the error state. For example, an error state may be associated with a known solution, or a solution derived based on error reporting communications received via the helpdesk service **632**. For example, the server agent **620** may train the AI model **628** based on an error state detected on the first user device **604***a*, accessibility data of the first user device **604***a*, and a solution to the error state implemented on the first user device **604***a*. In such instances, the server agent may later detect (e.g., using the AI model **628**), based at least on accessibility data received from the second user device **604***b*, the occurrence of the error state on the second user device **604***b*. Responsive to detecting the occurrence of the error state using the AI model **628**, the server agent **620** may output, based on at least the AI model **628**, a command to the endpoint agent **608***b* executing on the second user device **604***b* to implement the solution to error state on the second user device **604***b*.

[0129] The solution may include, for example, restarting at least one application executing on the second user device **604***b*, modifying an application configuration of at least one application of the second user device **604***b*, modifying a network configuration of the second user device **604***b*, and/or the like.

[0130] In some instances, the server agent **620** also provides endpoint management data to the AI model **628** as training data for detecting and troubleshooting error states. For example, the server agent **620** may train the AI model **628** using the endpoint management data received from the first user device **604***a*, the accessibility data of the first user device **604***a*, the error state of the first user device **604***a*, and the solution to the error state on the first user device **604***a*. Accordingly, the server agent **620** may later predict, using the AI model **628**, an occurrence of the error state on another user device (e.g., the second user device **604***b*) based on endpoint management data received from the second user device **604***b*. For example, the server agent **620** may determine that a particular configuration of the first user device **604***a* resulted in the error state, and that the second user device **604***b* has the same configuration. In such instances, the server agent **620** may output, based on at least the AI model **628**, a command to the endpoint agent **608***b* executing on the second user device **604***b* to preemptively implement the solution to the error state on the second user device **604***b*.

[0131] After outputting a command to implement a solution to an error state on the second user device **604***b* (e.g., responsive to detecting or predicting the error state on the second user device **604***b*), the server agent **620** may receive new endpoint management data from the second user device **604***b* and train the AI model **628** based on the new endpoint management data. The AI model **628** may therefore be trained based on the results of an implementation of a solution on the second user device **604***b*, such that the AI model **628** may further improve on known solutions to error states.

[0132] Referring now to FIG. **8**, an example method **800** for training the AI model **628** to detect error states is illustrated. The method **800** includes making (e.g., with the server agent **620** in conjunction with the first endpoint agent **608***a*) a first call via an accessibility API to request accessibility data associated with a first UI displayed by the first user device **604***a* (at block **804**). The server agent **620** receives the first accessibility data via the accessibility API (at block **808**), and trains, based on at least the first accessibility data and an error state associated with the first user device **604***a*, the AI model **628** (at block **812**). As described above, in some instances, the error state of the first user device **604***a* is reported through the helpdesk service **632**. Alternatively, or in addition, the server agent **620** is configured to determine the error state of the first user device **604***a* based on at least first endpoint management data received from the first user device **604***a*.

[0133] The method **800** further includes making (e.g., with the server agent **620** in conjunction with the second endpoint agent **608***b*) a second call via the accessibility API to request second accessibility data associated with a second UI displayed by the second user device **604***b* (at block **816**). The server agent **620** receives the second accessibility data via the accessibility API (at block **820**), and detects, based on at

least the second accessibility data and the AI model **628**, the error state on the second user device **604***b* (at block **824**).

[0134] As described above, in some instances, the server agent **620** further trains the AI model **628** (e.g., at block **812**) based at least on a solution to the error state. In such instances, the server **624** may output a command to an agent executing on the second user device (e.g., the second endpoint agent **608***b*) to implement the solution to the error state (at block **828**).

[0135] Referring now to FIG. **9**, an example method **900** for training the AI model **628** to predict error states is illustrated. The method **900** includes making (e.g., with the server agent **620** in conjunction with the first endpoint agent **608***a*) a first call via an accessibility API to request accessibility data associated with a first UI displayed by the first user device **604***a* (at block **904**), and receiving, with the server agent **620**, the first accessibility data via the accessibility API (at block **908**). The server agent **620** also requests and receives (e.g., using the first endpoint agent **608***a*) first endpoint management data from the first user device **604***a* (at block **912**).

[0136] The server agent **620** trains the AI model **628** based on at least the first accessibility data, an error state associated with the first user device **604***a*, and the first endpoint management data (at block **916**). As described above, in some instances, the error state of the first user device **604***a* is reported through the helpdesk service **632**. Alternatively, or in addition, the server agent **620** is configured to determine the error state of the first user device **604***a* based on at least first endpoint management data received from the first user device **604***a*.

[0137] The method **900** further includes requesting and receiving (e.g., with the server agent **620** in conjunction with the second endpoint agent **608***b*) second endpoint management data from the second user device **604***b* (at block **920**). The server agent **620** predicts, based on at least the AI model **628** and the second endpoint management data, an occurrence of the error state on the second user device **604***b* (at block **924**).

[0138] As described above, in some instances, the server agent **620** further trains the AI model **628** (e.g., at block **916**) based at least on a solution to the error state. In such instances, the server **624** may output a command to an agent executing on the second user device (e.g., the second endpoint agent **608***b*) to preemptively implement the solution to the error state (at block **928**).

[0139] Although specific embodiments have been described, various modifications, alterations, alternative constructions, and equivalents are also encompassed within the scope of the disclosure. Embodiments are not restricted to operation within certain specific data processing environments but are free to operate within a plurality of data processing environments. Additionally, although embodiments have been described using a particular series of transactions and steps, it should be apparent to those skilled in the art that the scope of the present disclosure is not limited to the described series of transactions and steps. Various features and aspects of the above-described embodiments may be used individually or jointly.

[0140] Further, while embodiments have been described using a particular combination of hardware and software, it should be recognized that other combinations of hardware and software are also within the scope of the present disclosure. Embodiments may be implemented only in hardware, or only in software, or using combinations thereof. The various processes described herein can be implemented on the same processor or different processors in any combination. Accordingly, where components or services are described as being configured to perform certain operations, such configuration can be accomplished, e.g., by designing electronic circuits to perform the operation, by programming programmable electronic circuits (such as microprocessors) to perform the operation, or any combination thereof. Processes can communicate using a variety of techniques including but not limited to conventional techniques for inter process communication, and different pairs of processes may use different techniques, or the same pair of processes may use different techniques at different times.

[0141] The specification and drawings are, accordingly, to be regarded in an illustrative rather than a restrictive sense. It will, however, be evident that additions, subtractions, deletions, and other modifications and changes may be made thereunto without departing from the broader spirit and scope as set forth in the claims. Thus, although specific disclosure embodiments have been described, these are not intended to be limiting. Various modifications and equivalents are within the scope of the following claims.

[0142] The use of the terms "a" and "an" and "the" and similar referents in the context of describing the disclosed embodiments (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. The terms "comprising," "having," "including," and "containing" are to be construed as open-ended terms (i.e., meaning "including, but not limited to,") unless otherwise noted. The term "connected" is to be construed as partly or wholly contained within, attached to, or joined together, even if there is something intervening. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein and each separate value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., "such as") provided herein, is intended merely to better illuminate embodiments and does not pose a limitation on the scope of the disclosure unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the disclosure.

[0143] Disjunctive language such as the phrase "at least one of X, Y, or Z," unless specifically stated otherwise, is intended to be understood within the context as used in general to present that an item, term, etc., may be either X, Y, or Z, or any combination thereof (e.g., X, Y, and/or Z). Thus, such disjunctive language is not generally intended to, and should not, imply that certain embodiments require at least one of X, at least one of Y, or at least one of Z to each be present.

[0144] Preferred embodiments of this disclosure are described herein, including the best mode known for carrying out the disclosure. Variations of those preferred embodiments may become apparent to those of ordinary skill in the art upon reading the foregoing description. Those of ordinary skill should be able to employ such variations as appropriate and the disclosure may be practiced otherwise

than as specifically described herein. Accordingly, this disclosure includes all modifications and equivalents of the subject matter recited in the claims appended hereto as permitted by applicable law. Moreover, any combination of the above-described elements in all possible variations thereof is encompassed by the disclosure unless otherwise indicated herein.

[0145] All references, including publications, patent applications, and patents, cited herein are hereby incorporated by reference to the same extent as if each reference were individually and specifically indicated to be incorporated by reference and were set forth in its entirety herein.

[0146] In the foregoing specification, aspects of the disclosure are described with reference to specific embodiments thereof, but those skilled in the art will recognize that the disclosure is not limited thereto. Various features and aspects of the above-described disclosure may be used individually or jointly. Further, embodiments can be utilized in any number of environments and applications beyond those described herein without departing from the broader spirit and scope of the specification. The specification and drawings are, accordingly, to be regarded as illustrative rather than restrictive.

What is claimed is:

1. An electronic device comprising:

at least one electronic processor configured to:

make a first call via an accessibility application programming interface ("API") to request first accessibility data associated with a first user interface ("UI") displayed by a first device;

receive the first accessibility data via the accessibility API, the first accessibility data comprising at least one of (a) information identifying one or more UI elements in the first UI or (b) information identifying one or more UI events in the first UI;

train, based on at least the first accessibility data and an error state associated with the first device, an artificial intelligence ("AI") model;

make a second call via the accessibility API to request second accessibility data associated with a second UI displayed by a second device;

receive the second accessibility data via the accessibility API, the second accessibility data comprising (a) information identifying one or more UI elements in the second UI or (b) information identifying one or more UI events in the second UI; and

detect, based on at least the second accessibility data and the AI model, the error state on the second device.

2. The electronic device of claim **1**, wherein the error state of the first device is reported through a helpdesk service.

3. The electronic device of claim **1**, wherein the at least one electronic processor is further configured to determine the error state of the first device based on at least first endpoint management data, the first endpoint management data including state information of the first device.

4. The electronic device of claim **1**, wherein the AI model is further trained based on at least a solution to the error state, and wherein the at least one electronic processor is further configured to:

output, based on at least the AI model, a command to an agent executing on the second device to implement the solution on the second device.

5. The electronic device of claim **4**, wherein the solution includes at least one selected from a group consisting of restarting at least one application executing on the second device, modifying an application configuration of at least one application of the second device, and modifying a network configuration of the second device.

6. The electronic device of claim **1**, wherein the AI model is further trained based on first endpoint management data and a solution to the error state, the first endpoint management data including state information of the first device,

wherein the at least one electronic processor is further configured to:

predict, based on at least the AI model and second endpoint management data including state information of a third device, an occurrence of the error state on the third device; and

output, based on at least the AI model, a command to an agent executing on the third device to preemptively implement the solution on the third device.

7. The electronic device of claim **6**, wherein the at least one electronic processor is further configured to:

receive third endpoint management data after outputting the command, the third endpoint management device including state information of the third device; and

train the AI model based on at least the third endpoint management data.

8. The electronic device of claim **1**, wherein the information identifying the one or more UI elements in the first UI includes at least one selected from the group consisting of: (a) a respective element type of the one or more UI elements in the first UI, (b) a respective identifier of the one or more UI elements in the first UI, or (c) a respective state or condition of the one or more UI elements in the first UI.

9. The electronic device of claim **1**, wherein the information identifying the one or more UI elements in the first UI is organized as a hierarchical tree.

10. The electronic device of claim **9**, wherein

the one or more UI elements in the first UI comprises a first UI element that includes a second UI element;

the hierarchical tree comprises a first hierarchical level that is above a second hierarchical level;

the first hierarchical level includes a first node representing the first UI element; and

the second hierarchical level includes a second node representing the second UI element.

11. The electronic device of claim **1**, wherein the information identifying the one or more UI events in the first UI includes one or more notifications of changes in states or conditions of the UI elements in the first UI.

12. The electronic device of claim **1**, wherein the first call made via the accessibility API is made to a platform rendering the first UI.

13. The electronic device of claim **12**, wherein the platform includes one of an operating system executing on the first device or a browser application executing on the first device.

14. The electronic device of claim **1**, wherein the first accessibility data is generated based on metadata associated with the one or more UI elements in the first UI, the metadata being exposed via the accessibility API.

15. The electronic device of claim **14**, wherein the metadata is specified in one of (a) application code of application for which the first UI is being rendered or (b) content code of web content for which the first UI is being rendered.

16. The electronic device of claim **14**, wherein the metadata specifies hierarchical relationships between the one or more UI elements in the first UI.

17. The electronic device of claim **1**, wherein the first accessibility data is generated by a platform executing on the first device during rendering of the first UI.

18. A method for training an artificial intelligence ("AI") model for error troubleshooting, the method comprising:

making a first call via an accessibility application programming interface ("API") to request first accessibility data associated with a first user interface ("UI") displayed by a first device;

receiving the first accessibility data via the accessibility API, the first accessibility data comprising at least one of (a) information identifying one or more UI elements in the first UI or (b) information identifying one or more UI events in the first UI;

training, based on at least the first accessibility data and an error state associated with the first device, the AI model;

making a second call via the accessibility API to request second accessibility data associated with a second UI displayed by a second device;

receiving the second accessibility data via the accessibility API, the second accessibility data comprising (a) information identifying one or more UI elements in the second UI or (b) information identifying one or more UI events in the second UI; and

detecting, based on at least the second accessibility data and the AI model, the error state on the second device.

19. The method of claim **18**, further comprising:

training the AI model based on at least a solution to the error state; and

outputting, based on at least the AI model, a command to an agent executing on the second device to implement the solution on the second device.

20. An electronic device comprising:

at least one electronic processor configured to:

make a first call via an accessibility application programming interface ("API") to request first accessibility data associated with a first user interface ("UI") displayed by a first device;

receive the first accessibility data via the accessibility API, the first accessibility data comprising at least one of (a) information identifying one or more UI elements in the first UI or (b) information identifying one or more UI events in the first UI;

train an artificial intelligence ("AI") model based on at least the first accessibility data, an error state associated with the first device, and endpoint management data associated with the first device, the endpoint management data including state information of the first device; and

predict, based on at least the AI model and second endpoint management data including state information of a second device, an occurrence of the error state on the second device.

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